STELLAR CYBER®

STELLAR CYBER OPEN XDR PLATFORM INSTALLATION AND CONFIGURATION

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L00 - INTRODUCTION

L00A-PRES: Introduction to Stellar Cyber Platform Installation (04m:15s)

00:00-01:00
Hi and welcome to *Stellar Cyber Open XDR Platform Installation and Configuration* module. I'm David Shirley from the Stellar Cyber Enablement and Training teams. The lessons in this module show how to install, configure, and verify a Stellar Cyber Data Processor in VMware's ESXi hypervisor. We will cover the configuration of ESXi data stores, ESXi networking, the creation and configuration of the Data Lake and Data Analyzer virtual machines, and the final configuration in the Stellar Cyber user interface. Throughout this module, there may be new concepts and terms that you may not be familiar with. But that's okay and to be expected. The important thing is that you become familiar with the concepts and the terms described. Throughout the Stellar Cyber Open XDR training curriculum, your understanding and application of cybersecurity principles...

01:00-02:00
...will deepen with each repetition, iteration, and example. As you advance, you're on your way to becoming a cybersecurity superstar. Thank you for your dedication.

Welcome to *Stellar Cyber Open XDR Platform Installation and Configuration*. The following lessons are included in this module:

- *High-level Platform Component Roles*. At a high level we will see deployment architectures and the role of the Data Lake and the Data Analyzer in a Stellar Cyber platform deployment.
- *ESXi Data Store Configuration*. This lesson starts with an overview of the resource requirements of the Data Lake and the Data Analyzer virtual machines followed by the creation of the Data Stores in ESXi.

02:00-03:00
- *ESXi Networking Configuration.* This lesson starts with an explanation of the basic cluster architecture and the dedicated cluster architecture followed by the networking requirements in ESXi.
- *Virtual Machine Creation - Data Lake (DL).* With the Data Store and networking complete in ESXi, next we will download the software image and create the Data Lake virtual machine.
- *Stellar Cyber Platform - Data Lake Configuration*. This lesson starts with powering on the Data Lake virtual machine and then the initial console CLI configuration for the Data Lake.
- *Virtual Machine Creation - Data Analyzer.* In this lesson, we create the Data Analyzer virtual machine in ESXi.

- *Stellar Cyber Platform - Data Analyzer Configuration*. This lesson starts with powering on the Data Analyzer virtual machine, followed by the initial console CLI configuration for the Data Analyzer.

03:00-04:15

- *Stellar Cyber Platform - Data Processor Configuration*. With the Data Store and networking in ESXi complete, and the creation and configuration of the Data Lake and Data Analyzer virtual machines complete. This lesson finishes the set up step in the Stellar Cyber user interface from first login to validation of the Data Lake, Data Analyzer, receivers, and sensor profiles.

- *Data Lake - Data Lake Additional Drive*. Now that the data processor is up and running, The final piece is to add the data store for the Data Lake to the Data Lake virtual machine and then migrate the data to this new large storage directory.

- *Extended Content - Reconfiguring the Data Processor*. This lesson demonstrates how to modify both ESXi and the Stellar Cyber Data Processor to separate the communication between the Data Lake and the Data Analyzer from the management and log forwarding communication.

Now let's get going and happy learning.

L00B-PRES: High-Level Platform Component Roles (08m:05s)

00:00-01:00

The first lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *High-Level Platform Component Roles*. *High-Level Platform Component Roles*. This lesson includes a high-level look at the platform components and their roles.

Stellar Cyber platform component for both on-prem and cloud consist of a data processor, which is made up of a Data Analyzer and a Data Lake. From a Stellar Cyber sensor perspective, there are modular sensors. They are both physical the photon sensors, and virtual appliances installed in a hypervisor, something like ESXi or Hyper V or KVM, then we've got two server sensors. We have the Windows Server sensor...

01:00-02:00

...where some Stellar Cyber software is loaded on the Windows Servers, and then the Linux Server sensor where Stellar Cyber software is loaded onto the Linux server. In a cloud environment, Stellar Cyber takes care of the back end so there's no need for you to be concerned about the Data Analyzer or the Data Lake, that is taken care for you in the SaaS deployment of Open XDR platform, but we still have the Stellar Cyber sensors, the Modular Sensor, the Windows Server sensor and the Linux Server sensor. The components: So the Stellar Cyber Open XDR platform for on prem and private-hosted cloud deployments if you had your own environment, your own AWS, or your own Azure or your own GCP, or your own Oracle Cloud OCI.

02:00-03:00

It's called the data processor. It's got two main components that we've already spoken about, installed as virtual machines. They are the Data Analyzer, and this includes the machine learning,

correlation, enrichment, and the cloud connectors, and then we have a Data Lake. It contains the user interface, the hot storage, warm storage, does the search, the application programming interface or API, and the Data Sink. The Data Sink can simultaneously sync data to the Data Lake storage and to the cold storage systems. So note Stellar Cyber platform is also available as we just described as a SaaS deployment where the architecture and the maintenance of the platform is managed by Stellar Cyber. So the platform deployment types, so this is for an on-prem deployment.

03:00-04:00
This is not a SaaS deployment. We can deploy either as a single node deployment with the data processor installed as a single Data Analyzer node and a single Data Lake node. So just one of each is a single deployment. For a more scalable environment, we can deploy in a cluster deployment with the data processor installed as a Data Analyzer cluster and the Data Lake as a cluster to achieve higher and ingest higher data ingestion rates, greater storage capability, and longer data retention in warm storage. The Data Analyzer cluster: It's a collection of Data Analyzer instances or virtual machines as we've been talking about that will work together to process and analyze the data. The types of Data Analyzer instances are designed as either master or worker nodes. So we have a Data Analyzer master–it's the node that coordinates the clusters and manages its operation–and then we have the Data Analyzer worker nodes...

04:00-05:00
...that execute the data processing tasks, and contribute to increased processing capacity and resiliency. The Data Lake instance: So they're designed as a master worker and coordinating node to work together to store managed data in the Data Lake. So the Data Lake master node oversees the storage management and the search operations, and then the Data Lake Worker nodes handle the actual storage. A coordinating node is a special type of Data Lake worker node that focuses on coordinating data searches and requests from different services to the data nodes in the cluster, improving our overall performance. So the advantage of the clustered architecture, we have improved performance so we can distribute the workload across multiple nodes for faster processing.

05:00-06:00
Scalability: So easily handle more nodes to handle the increased load of the data. So as the deployment grows we can continue to add worker nodes to increase that scalability. High availability: So redundancy ensures that the system remains operational, even if a single node fails we have built in redundancy with high availability. Workload distribution: So it manages more data and enhances the processing capabilities and some fault tolerance. So it continues functioning despite individual node failures relates back to our high availability. Okay. Stellar Cyber Open XDR platform for on prem and cloud has two main components. Virtual machine: So we talked about the Data Analyzer, which includes the machine learning, the correlation, the enrichment of Interflow records, and the cloud connector components.

06:00-07:00
So we have a data master or how we've written here D-A and a small m, and optional DA workers within the cluster to support extra capacity. When you first create this you don't have to have all of those workers available, but we do create it from a start to say it's going to be a clustered

environment The Data Lake contains user interface, Data Sink, and Elasticsearch components. So the Data Lake master or as a reference here DLm, and optional Data Lake workers within the cluster to support extra capacity. Just a note at the bottom there that Stellar Cyber platform is available as a SaaS deployment architecture where the maintenance of the platform is managed by Stellar Cyber. Alright. The cluster components we've talked a little bit about these. We have a DAm or a DA Master. That's the Data Analyzer master node that coordinates the activities of the worker nodes and manages the overall cluster.

07:00-08:05
Then below that data master we have the worker nodes, the Data Analyzer workers. They perform data ingestion from the sensors and connectors, as well as enrichment. It helps increase data processing capacity and improves fault tolerance. The Data Lake Master, here DLm, that manages storage and search operations in the Data Lake. It stores data itself and coordinates data searches and requests from different data processing services. But beneath that we have Data Lake workers. so that handles the actual storage of data in the Data Lake. They store the Elasticsearch data and can be designated as a coordinating node to improve overall Data Lake performance.

The lesson, *High-Level Platform Component Roles*, is complete. You should now be familiar with the Stellar Cyber Open XDR platform components and their roles. Thank you.

L01 - ESXi DATASTORE CONFIGURATION

L01A-PRES: ESXi Datastore Configuration (15m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *ESXi Datastore Configuration. ESXi Datastore Configuration.*

Let's start by looking at the minimum system requirements for our Data Processor including the Data Analyzer and the Data Lake. So we can see here, installation–and this example is for VMware's ESXi. So the installation of the Stellar Cyber Platform on VMware ESXi requires the following. So the Data Lake virtual machine needs forty CPU slash virtual CPUs, five hundred gigs of SSD provisioned thick, and the Data Lake storage. In a production environment, twelve or more terabytes of SSD disk, again thick provisioned.

01:00-02:00
And in a lab environment, our environment, or a test environment, at least two terabytes of SSD disk again, thick provisioned. The Data Analyzer virtual machine the DA requires 44 CPU or virtual CPUs and 500 gig of SSD. That's just for the virtual machine, again, thick provisioning. So let's look at the recommended platform requirements for our Data Processor. We can see here for the Data Lake highlighted, requires 40 CPU, 136 gig of RAM, and 500 gig of OS SSD disk space. And with the Data Lake attached to it, another drive of twelve or more terabytes. So you can install a DA and a DL virtual machines in the same datastore...

02:00-03:00
...if there is sufficient space for the virtual machines and the twelve terabytes required for the Data Lake's Elasticsearch data. However, Stellar Cyber recommends that the Data Lake use a dedicated datastore, so one datastore for the virtual machines and a separate datastore in ESXi for the Data Lake. So we can see there the host still needs to accommodate. So we can't use the entire number of CPU up for the Data Lake and the Data Analyzer. We must leave some CPU and some memory for the host to run–actually ESXi box to run. So the minimum requirements here So this is when we're looking at a test environment or a training environment. As we can see there we can install the DA and the DL. Those virtual machines in the same datastore if there's enough space, but we don't recommend you do that.

03:00-04:00
So at a minimum requirement so we can see the CPU has dropped. We're now down to 16 virtual CPUs. The RAM has not changed. Actually I think the RAM did go down from 136 to 128 and the disk space remains the same as 500 gig and a minimum of two terabytes for that Data Lake where Elasticsearch will be storing the data. So minimum platform requirements continued. Management IP address for the ESXi host and all virtual machines so if we're deploying this as a standalone without clustering, we only need one IP address. So one public IP address for each Data Processor for management access. So when we say Data Processor that includes both the DA and the DL. So just one for there. But if we deploy it in a clustered environment, we need two public IP addresses for each of our DPs in a cluster.

04:00-05:00
One for the management access and one for the cluster itself, and we'll see where those two addresses are used. From a firewall component so a reachability perspective of how do we reach the Data Processor and how does the Data Processor reach the services it needs. So if we go to our documentation, and search for firewall requirements, you will find that there is a table. Wherever you see DA or DL, and then either as a destination or as a source to make sure that the ports for those protocols are opened if there is a firewall between the devices that are looking to communicate. Alright So let's do the storage So we're starting to talk about datastores in ESXi. So the Stellar Cyber platform is the Data Processor. Contains the DA Master virtual machine, and the Data Lake virtual machine the Data Lake Master virtual machine.

05:00-06:00
So two datastores are created or modified in this instance with the correct size. So one terabyte for our Stellar Cyber platform Data Processor, those virtual machines, one virtual machine for the Data Lake and one virtual machine for the Data Analyzer, and then we'll need a second datastore for our Data Lake storage where the Elasticsearch will go. So in lab environments we saw before with those minimum requirements, that at least two terabytes can be used for that. So let's go ahead and see how to actually create this in ESXi. So the first one we're going to create here, as you can see, is the name the Stellar Cyber virtual machines. This is where the Data Lake virtual machine and the Data Analyzer virtual machine will be stored. So we give it a name and choose an available disk, and select **Next**.

06:00-07:00
Now when we do the partitioning, we will start with a full disk. We want to use the entire disk of just one partition and it will show you there what the before partition looked like and what the after partition looked like because this drive hasn't been used before. There is no before partition. And the other thing you've got to choose is which virtual machine file system will it use depending on your version of the ESXi. I would always recommend that you choose the latest VM file system for better performance and scalability. So choose **Full disk** and the latest version of VMFS and select **Next**. Alright. So that's all required to create our first datastore. So we look at the summary here and if everything looks okay there we select **Finish** to complete that setup. Alright. Now we want to add to that.

07:00-08:00
So we decided it's not large enough. We want to add an extent to an existing VMFS datastore. So one of our datastores we want to add to that. So we choose **Add an extent to existing VMFS datastore** and select **Next**. We will choose which datastore that is that we're going to add an extent. In this example, it's our Stellar platform VMs. That's where our Data Lake and our Data Analyzer will go and select **Next**. From there, we select a disk. We can only choose one at a time So from the available disks we select **Next**, choose one and sorry choose one and select **Next** to add that hard drive. In this example you can see they are SSD disks to add that to the datastore. Again it will ask you do you want to use the entire disk for the extent or partition of that disk.

08:00-09:00
As we can see here that before we select partition is hasn't been used before so there is no partition. And afterwards it will look like one large VMFS partition that will be added to our datastore. So select **Next**. We'll be given a summary of what we're about to do that we've chosen the datastore Stellar platform VMs. We can see the disk that we've selected that we're going to partition the entire disk and the VMFS version is number six for this. So we select **Finish** to continue, but will pop up with a warning saying, do you know that you're going to override whatever is on that disk? And of course we're okay with that because there's nothing on that disk. It's a new disk, so I will select **Yes** to continue. Alright. So we can see here our disk has grown in size. It now has two terabytes of capacity where previously it only had one terabyte. So we've added that second disk to have a greater size datastore for those virtual machines that will sit there.

09:00-10:00
So we select **Storage** and **datastores** and you can see that our VM has increased. Alright. Now we're going to create another datastore. So we select **New datastore** to create the datastore for our Data Lake. Not the Data Lake virtual machine that will go on this Stellar platform VMs, but another datastore where our Elasticsearch data will actually be stored. Alright. We're going to create a new VMFS datastore. That's the selection we make and we select **Next**. Give it a name In this example it's SC Data Lake Storage, and we have to select a disk. You can see that the list is starting to shrink as we're using up the disk. We used two in that previous datastore.

10:00-11:00

Here is the beginning of the creation of this datastore by selecting one of these drives. And you can see as we continue in this module, and add more drives to this datastore we can see that the list will shrink as they get used up. So give it a name, select the disk, and choose **Next**. Again we'll be asked the question, do we want to use the entire disk? And that's a yes and what VMFS, what file storage system in this version of ESXi 7.0.3 VMFS 6 is available, and we choose that and select **Next**. Again same as our Stellar Cyber virtual machine datastore. This one is going to ask you the question. Here is a summary of what you've got. Click **Finish** if you're happy with that. And we'll see now again we'll get that pop up just to make sure to let you know we need a confirmation that you are about to overwrite any data that are on those disks. And if that's okay we select **Yes** to continue.

11:00-12:00

Okay. So now we have two datastores. We have our Stellar Cyber platform VMs. That's the first one we created, and then we added a second disk to it. So it's now got two terabytes of storage, and we've just newly created the Stellar Cyber Data Lake storage datastore. Now we've got two of them. You can see the capacity of the SC Data Lake storage is only 1.75 terabytes, so certainly not enough for even a test environment or a demo environment and certainly nowhere near enough for a production environment. So let's go ahead and increase the capacity of the Data Lake store. We do that by selecting the name of that. We have this menu that we can see here. So along the top we don't want to register a VM. We don't want to browse the datastore. We don't want to refresh or we don't want to do the actions we want to select **Increase capacity**.

12:00-13:00

This is where we'll start choosing drives to add to this datastore. Select increase capacity. It could ask you how would you like to increase that capacity? and we're going to add an extent to an existing VMFS datastore. So we choose the top option and select **Next**. Here is our list of disks that are available to choose from to add to that datastore and remember they must be SSD disks. So if there was a selection of disks available on this host we're going to choose our SSD disks to add to this datastore. So choose one and select **Next**. Okay. It's going to ask you what sort of configuration do you want for this disk. We're going to use again the entire disk in a VMFS 6 format, and we're going to then select **Next.**

13:00-14:00

We'll be given a summary of what we're about to do. So for the SC Data Lake storage datastore, we're adding that disk and with a partition of using the full disk and a VMFS version of number 6. We'll be given a last chance to say, “Are you sure we're about to erase all of the content that is on that disk?” And if we are sure we select **Yes** to continue. Alright. So here's a summary of what we've got. Now we can see here that we now have two extents or two disks have been added: extent zero and extent one. They're the first two we've added. One was when we created, and then we just added the second one previously. We're going to keep increasing that capacity. Just continue to go through those steps until the datastore is the capacity that you need it to be. It's just a matter of repeating those steps.

14:00-15:00

So select the datastore, click **Increase capacity**, and then select a drive. Choose the full disk, choose the VMFS storage type, and then continue. So we've been around in circles and around in circles. We didn't show you all of the individual pieces, but you can now see after adding disks and adding disks we can now see a capacity of more than twelve terabytes. So for a production network or production deployment of our Data Processor, this is a great result. We have datastore Stellar platform VMs where we will host our Data Lake VM and our Data Analyzer VM. Remember, they both needed a minimum of 500 gig of disk space, so two terabytes. There's plenty of space, nearly more than double available for that.

15:00-15:24
And for the storage of our data, we have a different datastore SC Data Lake storage and we can see for a production environment of more than twelve terabytes that is ready to go as well. So all complete here from a datastore configuration in ESXi. The lesson ESXi datastore configuration is complete. Thank you.

L01B-DEMO: ESXi Datastore Configuration (18m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *ESXi Datastore Configuration. ESXi Datastore Configuration*. In this lesson, we'll cover the system requirements for installing a Data Lake and Data Analyzer virtual machines in VMware ESXi. We'll also create two datastores in ESXi, one for the virtual machine images, and the other for the Data Lake storage.

Let's start with the system requirements for installing a Data Lake and Data Analyzer virtual machines in VMware ESXi. Let's go to the documentation for that and I'll show you where it is. So here we are. First thing I want to point out is that we are looking at the 4.3.7 documentation.

01:00-02:00
5.1.1 for on prem is available. But at this stage, what we want to do is install 4.3.7 so we have the opportunity to show an upgrade in a later lesson. We'll start with 4.3.7. For the install of a Data Processor on ESXi, we'll go to **Installing**. Hit the drop down. (Select) **Installing a Data Processor**, hit the drop down. You can click on the link and it will just populate the page. It's just a little slower doing it that way. I prefer to go down here. The one we're going to do is (titled) *Installing a Data Processor in VMware (Cluster)* as opposed to installing an all in one where it's bundled together into one VM image. We're going to do the more scalable installation in VMware where we have the option of either doing a basic or a cluster installation. We'll click on that.

02:00-03:00
Before you begin–there's a couple of before you begins and we want to talk about that. A dedicated ESXi server running eight seven or six dot seven. We are running. If we go to **Help** and **About**, we are running 7.0.3, ESXi version 7.0.3. We fit with our 7.0 right there. 6.7 is the oldest one we'll support and as we know ESXi 8.0 is the latest ESXi available right now. Okay. We'll go through these. There are a couple of requirements depending on the installation. If we do a non clustered or it's often called a

basic installation, we only need one public IP address to access the Stellar Cyber platform. If we do a clustered environment, we will need two public IP addresses.

03:00-04:00
We're going to install this in a clustered environment to start with. We're going to install the basic configuration where we have two VMs: a Data Lake and a Data Analyzer. It just comes down to–we'll go through that description–it just comes down to how many virtual switches we have. What that means is, "Do we have a virtual switch for communication between the Data Lake and the Data Analyzer?" or, "Is the Data Lake, Data Analyzer communication happening on the same virtual switch as the ingress of data from the network?" whether log forward data or network traffic data and of course management of the platform also will come in through that virtual switch. I'll specify that out a bit more as we go through this configuration. Okay. Ports are particularly important. What ports do we need to open our firewall if we click on there? Let's have a look. What we're interested in is the DA, the Data Analyzer...

04:00-05:00
...and the Data Lake, and the DP. They're all the ones we need to look at when we talk about this down the bottom. We'll go look in this column. We've got our source DA DL, And this is what is it communicating to? We've got a DL communicating to a mail server. Our data platform is able to send emails whether that's from an ATH playbook, which we'll cover in another lesson. We'll need to have these ports open for that to communicate a DA to who is needs port 43 on TCP. Then we come further down to an OKTA server: Where does that need to go to? NTP server: Where does that need to go through? Port twenty three UDP. There's our firewall ports that we need to have open for our data on to be able to communicate. Minimum requirements are very important. Okay. We've got a couple of requirements here. The first version of the requirements is a production environment.

05:00-06:00
You can see this is recommended for production. What we need here–we can see we've got resource of CPUs virtual CPUs, what's going to happen on the host to that host? Meaning our ESXi host, 44 physical or 88 cores with hyper threading, the DL will require 40 of those, and the DA 44 of those. From a RAM perspective, the Data Lake 136 gig, the Data Analyzer, 64 gig. Then the disk space when that OS is for the operating system, there are VMs. It was all going to have a virtual machine for the Data Lake and a virtual machine for the Data Analyzer. We can see there we need one terabyte of data. Five hundred gigs each: 500 for the DL and 500 for the DA. Okay. Moving on to the Data Lake disc space. You can see there we need 12 terabytes for a production version of the Data Lake.

06:00-07:00
Now if we go a little further down in this table, we can see minimum configurations for separate DP VMs. What we mean by separate is a separate Data Lake and a separate Data Analyzer virtual machines. We can see there at the low end, we can have 16 CPUs for the DA, 16 for the DL. RAM we still need a lot of RAM for that Data Lake 128 gig, and 64 gig is the same for the Data Analyzer, but what has not changed at all is the operating system space for our virtual machine. They're still 500 gig each. But the big change is the Data Lake. Now this would be if you were doing this in as we are, a demonstration environment or a training environment, you can see the Data Lake needs at least two

terabytes for the Data Lake. We'll actually configure it with a little bit more than that, but we probably won't go up to the 12 that's required even though we've got the space.

07:00-08:00
We'll add a few disks too, at least certainly get it over the two terabytes. Okay. The next thing we need to do we've of course we've installed VMware ESXi. Let's go on a little tour of that just now. As we saw before we're running 7.0.3. We can go to **Help** and then **About** and see that we're running 7.0.3. This has only just been created. If we have a look we can see this is a–it's got 44 CPUs. If we expand that out we'll see that logical processors– there's actually 88 of them. Memory–there's a heap of memory–256 gig of memory, and there is a bunch of storage in here. Let's have a look at what we've got. Virtual machines: No virtual machines so far. We still must create that Data Lake virtual machine and a Data Analyzer virtual machine.

08:00-09:00
Storage: At the moment there is only one datastore. This is being called the HDD datastore. This was a non SSD drive. I don't know if I mentioned that earlier, but all Data Lake and OS versions must run on SSD drives. We must have that speed for input output. So this datastore, we won't use this for the DL or the DA or certainly not for the Data Lake storage. This is just a non SSD. We'll create two datastores: one for the virtual machines, and we'll create a second datastore for the Data Lake. If we actually go and have a look at the devices we can see there's a bunch of devices in here that are ready to go. Here's all of our SSD disks. We've got 1, 2, 3, 4, 5, 6, 7, 8 of them. They're all 1.75 terabytes. We'll probably use...well we only need one of these for the VM storage. Remember they were 500 gigs each. We need at least two of them to get over the two terabyte limit for the Data Lake storage.

09:00-10:00
Maybe we'll do three or four of them just so you can see what that process looks like. From a networking perspective, let's have a look at what we've got. A bunch of NIC cards on the physical machine. Virtual switches: just what's come standard with it. We can see the standard vSwitch0 is there by default. There's one virtual switch and the port groups that are connected to that. We've got a VM network, and we've got a management network. Again, two default port groups that are using vSwitch0. Okay. We're going to configure that storage. Of course we're going to increase capacity and walk through how to do all that. Here's the documentation that will walk you through that, creating the datastores, verifying the datastore's discs, and then actually going ahead. We won't go into the actual console to do that.

10:00-11:00
What we want to do is then deploy our virtual machines: our DL master and our DA master is what we want to do. We've talked a little bit about it before–the difference in deployments. We're going to do a basic deployment, which just means there's one virtual switch. We'll create a virtual switch to do that or maybe we'll use vSwitch0. It's not important. Perhaps later we will upgrade this deployment to a cluster environment so we can see what that looks like with having a second virtual switch. You can see that down there. We've got a second virtual switch or a cluster virtual switch there. Let's go ahead and create those two datastores. Back to storage. Select **Storage** over here. We can see we had a

bunch of devices available. We have just the one datastore. We're going to create two more right now. Let's go ahead and do that. Alright.

11:00-12:00

To do that we click on **New datastore**. The first thing we're going to do is select **Create new VMFS datastore**. That's what we want to do. We're not going to add because we don't have anything to add and we're not going to mount an NFS one, we're going to create it on this device. Let's create the first one, the smaller one. Let's create the datastore that's going to house the virtual machines. We choose **Next**. Give it a name, and our name can be something like, how about that Stellar Cyber VMs. We can use that as our name for this datastore and we choose a disk. We said there were eight of them to choose from–one, two, three, four, five, six, seven, eight SSD disks all 1.75 capacity. We'll just choose the first one. There's no need to pick one or the other here. Select **Next**. Now it's going to ask what's going to happen here. We want to use the full disk. So maybe it was partitioned before but these are fresh disks so there's no partitioning...

12:00-13:00

...and which file system this VMFS stands for VMware file system. You can see we have a choice between five and six. We'll choose the latter one that has better performance. Select **Next**. It gives us a summary of what's going to happen. We're going to create a VM machine called Stellar Cyber VM. It's going to use that disk. It's going to use the entire disk and the virtual machine file system. I think I called it VMware before, the virtual machine file system version is number six for this installation of a datastore. Then click **Finish**. We'll get a confirmation here that will say we're about to wipe this disk. Are you sure you want to do this? Yes, we are. Okay. That's going to go ahead. We can see down at the bottom. Wow! That was really quick. Create the VMFS datastore. It just does a quick rescan, all successful, and we are ready to go. We can see up here we now have a datastore that says Stellar Cyber VMs. It's an SSD datastore with 1.75 terabytes, which is plenty for what we need.

13:00-14:00

We only needed 500 gigs for each of those virtual machines. Now let's create the next one, the datastore that we're going to put our Data Lake storage on. Again, (select) **New datastore**, (and then select) **Create a new VMFS datastore**. (Select) **Next**. We're going to call this one, why don't we call this Stellar Cyber Data Lake. Let's call it that. You can see now we've only got one, two, three, four, five, six, seven. There's only seven disks left. I probably didn't have to count them. I could've used the seven items over there. We'll just select one of them, and then select **Next**. Same as before–use the entire disk and virtual machine file system number six. Select **Next**. There's our summary.

14:00-15:00

Our datastore name is going to be Stellar Cyber Data Lake, the disk gets chosen. The partition is to use the entire disk, and the virtual machine file system version is number six. Select **Finish**. We'll get that same confirmation: You sure you want to wipe the disk? Yes, we do. Off it goes and creates that. Great. Now we've got two datastores, but we remember before the capacity for the Data Lake storage of 1.75 is not enough. We need to add some more disk to do that. The way we would do that is select the datastore. We're going to select Stellar Cyber Data Lake. There's a button up here

Increase Capacity. Select that. We're going to choose to **Add an extent to an existing virtual machine file system datastore** or VMFS datastore. Select **Next**. It's going to go look to see what's available. Now we're down to six.

15:00-16:00
We're going to choose one of those disks. Next, same as before: full disk, VMFS number six. Select **Next**. There's our summary. Select **Finish**. With a confirmation we're about to wipe it. Now it's going to add that second disk to the Data Lake datastore. Off it goes down here. We can see down the bottom: Extend VMFS datastore, Completed successfully. If we go back to our storage, we can see that our Data Lake now has a capacity of 3.49 terabytes. That's enough. That's more than the two minimum, but just for safekeeping and just to do it one more time, let's open up that Data Lake and we'll just add one more. Same again. Select **Add an extent to an existing VMFS datastore**. Yes. That's what we're doing. Select **Next**. Here comes the list of disks available. We're down to five. Just choose the top one. (Select) **Next** to continue.

16:00-17:00
Same again. Use the entire disk with that file system. What it would look like before, what the petition would look like after You can see all free space here. Afterwards it's all going to be used by VMFS. Next, there's our summary, (select) **Finish**. Yes, we want to wipe that disk. Confirm. Now back to our storage. We can see our Data Lake has now grown to 5.24 terabytes, so there's plenty of space for our training environment. If go in and really look at the details of this, it'll show you the three extents here. Are those three physical disks that are being used to create that storage. Just to confirm, we had the original HDD datastore. That was a non SSD drive, 500 gigs. We're not going to use that.

17:00-18:05
We created the Stellar Cyber VMs datastore. That's where we will put the virtual machines for a Data Lake and a Data Analyzer. Then we created another datastore: Stellar Cyber Data Lake. That's where we will create the storage for the Data Lake information will be stored there. We'll go see what this then looks like once we boot these machines up and configure them and see how we then include the use of that Data Lake storage in the DL VM. We'll see that through the system UI.

That's it for creating our ESXi for the moment, our datastore.

The lesson *ESXi Datastore Configuration* is complete. You should now be familiar with the system requirements for installing Data Lake and Data Analyzer virtual machines in VMware ESXi, as well as how to create datastores in ESXi.

L02 - ESXi NETWORKING CONFIGURATION

L02A-PRES: ESXi Networking Configuration (07m:35s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *ESXi Networking Configuration.*

ESXi comes with some default network configuration that we can use. And then there's some additional configuration that we'll talk about that we'd like to configure in ESXi for our deployment. Depending on what type of deployment you have, whether it's a basic architecture deployment or a clustered architecture deployment, will depend on what configuration steps we need to do within ESXi. Let's go ahead and look at what the requirements are and then how to do that configuration. By default, vSwitch0 is a virtual switch in ESXi that is there from default.

01:00-02:00
When you first configure ESXi on a box, on a host, the vSwitch0 will be there. We can use that for the management network to access the Stellar Cyber platform. You could, as the note says down below, you could optionally create a new vSwitch for the management network instead of using vSwitch0, but you can just use vSwitch0. There is a port group that we could use. We can either create a new port group or we can use a port group. What we want to talk about here is that we have our Data Processor made up of a Data Analyzer and a Data Lake. They both need to have a network interface. Those virtual machines will need to have a network interface so that they can communicate. In this diagram they're going to communicate via the vSwitch0, the virtual switch inside of ESXi. We can see there is a physical NIC: vmnic0 that is bound to vSwitch0, and that allows us to communicate with devices outside of the ESXi host.

02:00-03:00
Devices like connectors or modular sensors. To bring information in from the physical network, they'll come in via that management switch via a physical network adapter. In this case, vmnic0 into our virtual switch called vSwitch0, that then will have two interfaces, two port group interfaces, from the Data Analyzer and the Data Lake so that all that communication can happen. In a clustered environment, so we have a dedicated networking component for a clustered environment, we can see here vSwitch0 is still going to be used but this time it's only going to be used to bring data in and to manage the Data Processor.

03:00-04:00
For the DA and the DL, the Data Lake and the Data Analyzer, for those two virtual machines to communicate, in this example there is an additional virtual switch that's been created. This one's called vSwitch-Cluster. It's just a name. You can call it whatever you like. A new port group that the Data Analyzer and the Data Lake will use. And that network diagram it's called Stellar Cluster Network. And you can see that is attached to an outside physical switch via vmnic1. All of the configuration at the top of this diagram remains the same with vSwitch0 and vmnic0 and the port group of VM network. But below that in the bottom half of that diagram we've created an additional vSwitch, an additional port group with another physical NIC bound to that virtual switch to talk to that cluster switch.

04:00-05:00
Now on that cluster switch in the previous lessons we talked about having worker nodes for the Data Lake and a worker node for the Data Analyzer and coordinating nodes for the Data Lake. They could sit off that cluster switch on another ESXi host. That's what gives us our scalability and redundancy in

there. They all need to communicate and in this example they would communicate via a separate network as opposed to what's coming in through vSwitch0, which will be management of the data platform and our security and networking data coming from those devices either from log 14 or from connectors they will come in via that management switch. We can see here in the left side of our slide we've got a switch vSwitch0, which is by default and we're going to use that for the management of the device. We've got a vSwitch cluster for the traffic between the Data Lake master and the Data Analyzer master...

05:00-06:00
...and the cluster worker nodes and maybe a coordinating node as well for the Data Lake. We're going to create two new port groups. There's a port group for up the top. We can see it's port group VM network, and the bottom part of that network, for VM Switch cluster, we will have a port group called Stellar Cluster Network. Effectively in this architecture, the virtual machines will have two network interfaces where in the previous one they only had one network interface. How do we go ahead and create this? Okay. The first thing they're doing here is creating the port group in ESXi. The Stellar management network is what's being created here. We can see we access the UI of ESXi. We select **Networking**, select the **Port Groups** tab, and select **Add port group**.

06:00-07:00
This pop-up window will appear. We've got to give it a name. Stellar management network. You could call it SC management network. It's just a name. That's just the name of the port group VLAN. We don't need a VLAN ID in this but what's important is the virtual switch that's been selected. Now by default, the ESXi host only has one virtual switch, which is vSwitch0 and that's what's been selected here. We type in a name, we choose the virtual switch, and select **Add**. Then you'll see a summary of what was of the port group you've created. We can see there that we are connected to vSwitch0 and you can see that physical adapter connected in there vmnic0. Alright. Our port group now we have Stellar management network as opposed to the default port group, which is management network.

07:00-07:35
I say default as in that's what comes with ESXi. We've got our management network, for VMware ESXi management. We've got a Stellar management network and we're going to use that to access the Stellar Cyber platform. Then we're going to create a Stellar cluster network, and that will be used for internal communication between the Data Analyzer and the Data Lake.

The lesson, *ESXi Networking Configuration*, is complete. You should now be familiar with ESXi networking configurations. Thank you.

L02B-DEMO: ESXi Networking Configuration (10m:00s)

00:00-01:00
The next lesson in the *Open XDR Platform Installation and Configuration* module is *ESXi Networking Configuration. ESXi Networking Configuration*. In this lesson, we'll configure the ESXi networking component, the virtual switches, and the port groups acquired for the DP install of a Data Lake and the Data Analyzer.

Let's start with the configure, the networking components, vSwitches, and port groups. Alright. First of all let's look at let's go back to our documentation. We are doing 4.3.7, (select) **INSTALLING**, and then we're installing a data processor, and we're installing a data processor in VMware (cluster). Now even though it says cluster, we'll describe what we mean by the difference of those two.

01:00-02:00
We have these. We have basic deployment, all VMs on the same port group. We've got cluster deployment where we can see separated cluster management and SPAN traffic. Let's ignore SPAN traffic for the moment. We covered that in Modular Sensor in a different lesson. Let's just focus on our DL and DA and of course the management of the platform. If you click on these graphics, they'll expand them out. I'll just describe what we're seeing here. What we're seeing here is vSwitch0. We saw that in the previous lesson. That is the default switch in ESXi. We'll go into ESXi and do all of this. We can see we have the Management Network. That is the name of the port group that is attached to this virtual switch. And then it's also attached to VM NIC0. It may not be vmnic0. It could be a different network interface card...

02:00-03:00
...but this is a physical interface card that goes out to an external switch where there could be modular sensors, there could be network traffic, there could be log forwarding traffic will come in via this switch into our physical NIC that's connected to our virtual switch zero, which has a port group that's called Stellar Management Network. That interface–the port group–is used by the Data Lake Master, the Data Analyzer Master. Let's just ignore this SDS for the moment. That's the equivalent of our Modular Sensor. That's what they mean by the SPAN traffic. What we're seeing here really is that the Data Lake and the Data Analyzer are using the same virtual switch. They're actually also using the same port group in VMware to communicate to the utilizing vSwitch0...

03:00-04:00
...the same switch that then goes out to the network. That's the basic deployment, and that's what we're going to do to start with. Then we'll upgrade it in another lesson or maybe at the end of this lesson, to the next version. Let's have a look at that. The next version of this deployment is this cluster deployment. If we look at that, we can see that there is a vSwitch for cluster with a Stellar Cluster Network as a port group. So just think as one as an interface and one is a switch. Let's ignore the vSwitch SPAN. This is all talking about using it if we had an integrated meaning that there is a Modular Sensor also in our network design. We've already seen what that looks like in a previous lesson. Let's look at what this network looks like. We've got the same thing. We've got a switch coming in. We've got an external switch connected via vmnic0. That's a physical port that comes into virtual switch zero.

04:00-05:00
That has the port group Stellar management network that talks to our Data Lake master, our Data Analyzer master. If we were to manage it via the UI, that's the interface we'd come in through. So if our laptop was out here plugged into our switch coming through the physical interface into vSwitch0, which is our virtual switch. We arrive at our Port Group. It is called our interface that's called Stellar

management network. It's really unfair to call it Stellar Management Network. It's not really a network. This is a port group in this example. That's how we communicate to our DL Master and our DA master. But the difference here is the communication between the DL and the DA. There's a lot of communication that happens between the two, the Data Lake and the Data Analyzer, happens on another virtual switch.

05:00-06:00
This is a virtual switch called vSwitch-Cluster. Just like we've got vSwitch0, which is the default virtual switch in ESXi, another switch has been created called vSwitch-Cluster, and that allows these two devices, the DLm and the DAm these two virtual machines, to communicate on a different virtual switch. All of the communication that's happening there for a more scalable environment is not happening on the same interfaces that the data is arriving. The actual security and network data is arriving into our data platform. They're communicating on a different virtual switch with a different port group there. Now the reason that this one then has an external for itself is like, well if we've got a DL master and a DA master and they're very happy communicating over their shared port group, on their own virtual switch called vSwitch cluster, why would we need an external?

06:00-07:00
Well, if we think about expanding this where we have multiple DA VMs and DL VMs, we start talking about worker nodes in a much bigger environment where we scale, those other virtual machines may well be in an entirely different physical ESXi box. They all need to talk to each other. We would have an external switch with an entire another deployment of ESXi with a virtual switch with VMs and port groups to allow all of them to communicate where we can have a much more scalable environment. That's the reason for having this external portion of this picture out to another network. Okay. That's the theory on it. Let's go and set this up. Let's look at least what we have set up in ESXi. Okay. There's our storage.

07:00-08:00
We're all okay then. Nothing to do there right now. We can collapse that down. We're interested in the networking on this host. From a networking perspective, let's start at the bottom. We've got some physical NICs. We can see that vmnic0 is the only one of these NICs that's plugged in and is up running at a thousand megabits or one gig full duplex. We'll see where vmnic0 is bound. Let's look at virtual switches. We have one virtual switch. vSwitch0. If we go look at vSwitch0, we get a bit of a network diagram here. We can see that the topology here is that the physical NIC, vmnic0, is attached to this virtual switch called vSwitch0. We have at the moment two port groups attached to this virtual switch.

08:00-09:00
We've got VM network, and we've got management network. We're going to leave those two alone, and we're going to go create our own port group on this. If we go back to networking and port groups, we can see there are those two port groups: VM Network and management network. They're both using vSwitch0. We're going to create an additional port group called Stellar Cyber Management. We're going to use that interface for our basic deployment. Remember we're going to start off doing this basic deployment. We can use vSwitch0. We're going to create this port group,

Stellar management network, and that will be the interface that will be used for those two virtual machines. Let's go to **Add port group**. We said we would call it Stellar Cyber management. The only thing we have to choose is what virtual switches are connected to.

09:00-10:00
Well, there's only one configured so far: vSwitch0. That's it. Select **Add**. That will go ahead. We'll see it down at the bottom, updating network configuration. It's going to refresh it all happily. Completed successfully and we can now see we have Stellar Cyber management, port group, bound to vSwitch0. We are ready now to go ahead and build our VMs, our Data Lake VM and our Data Analyzer virtual machine. Let's go do that.

The lesson, *ESXi Networking Configuration,* is complete. You should now be familiar with configuring the ESXi networking component, the virtual switches, and the port groups acquired for the DP install of a Data Lake and a Data Analyzer. Thank you.

L03 - DATA LAKE VIRTUAL MACHINE CREATION

L03A-PRES: Data Lake VM Creation (10m:50s)

0:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Virtual Machine Creation Data Lake. Virtual Machine Data Lake*. In this lesson we will download the software image and create the Data Lake virtual machine.

Let's look at some of the requirements for the Data Lake and in this case the Data Analyzer of what we need to do to prepare for the creation. We've already prepared our ESXi host with datastores and with networking. We've created a datastore for the virtual machines. We've created a datastore for the Data Lake for the data of the Data Lake. We've created some networking infrastructure for both of the virtual machines.

01:00-02:00
Okay. To prepare, we talked about opening the ports on your firewall to their requirement. This image is brought straight out of our documentation. Contact Stellar Cyber Support for login credentials and your one-time password, which is also known as the license key. The credentials will allow us to download the software required and the OTP, or the license key, is what will license the software once we install it. So we will need that. Then we're going to download with the credentials we've received from support, an OVA file. That OVA file is used to install the same file is used to install our Data Processor. I mean by same file, the same file will be used for the Data Analyzer as would be used with the Data Lake. Now we can see there, there's our link, our https://acps.stellarcyber.ai/release.

02:00-03:00
In this example, we've got 4.3.7, but if you were trying to install either a previous or a later release, you just replace the 4.3.7 with 5.1.1 if you were doing 5.1.1. In the example we use in the demonstrations of this we've intentionally used 4.3.7 so we can show the upgrade process to 5.1.1. In this example, we're going to use 4.3.7 and download that OVA file. An OVA file is a file that has a bunch of components already in it that's ready to be deployed in our VMware ESXi environment. To do that, we navigate to get to this image. We navigate to the Knowledge Base and then go to **INSTALLING | Installing a Data Processor**, and then **Installing a Data Processor in VMware** in a cluster environment. We go there to locate this download link and we pull that down.

03:00-04:00
Now that we've got our software, we go into the user interface of ESXi to create the virtual machine. The first one we're going to create here is our Data Lake. We've got the networking that's already been configured. We have the datastores that have already been configured. Now we're configuring the virtual machines. In this instance from the left panel we choose **Virtual Machines**, and then from the top we select **Create / Register a virtual machine**, and we will get the next pop up. Okay. The pop up. How are we going to create this virtual machine? Well we've downloaded an OVA file. We're going to take the second selection of **Deploy a virtual machine from an OVF or an OVA file**. As I said we have an OVA file, so that's the selection. Choose the second one and select **Next**. Okay.

04:00-05:00
We've got to give it a name so we enter a name for the Data Lake VM. In this example, we are going to call it sc-dp, for Stellar Cyber Data Processor Data Lake Master, s c d p d l master, and then we've got to get that OVA file there. So you've got a choice. You can either drag that OVA file from wherever it is on your host to that blank space or if you click in that blank space it will open up a dialog box for you to browse to where you can locate that OVA file. Click Okay and then this is where we'll end up with giving it a name and we can see that our Aella Data Processor. So it's not specific to a DA or a DL. It's the same file used for both. And it's the same file whether it's a master or a worker node or a coordinating node. It's Aella, Data Processor, and then the version number. In this instance it's 4.3.7 and select **Next**.

05:00-06:00
Right. We're creating our Data Lake virtual machine. Where are we going to put it? We have got to choose one of our datastores. We've previously created these datastores. We created Stellar platform VMs. That's the one with two terabytes of data. Remember the requirement for this virtual machine is 500 gig of disk space. With two terabytes available in that datastore, that's plenty. The only other virtual machine we're going to add to that datastore is the Data Analyzer. Between the two of them it should be one terabyte. We've got double

the space available. The other datastore we can see there is the SC Data Lake storage, and that is used for the Data Lake's actual storage, not the VM storage. This one's talking about where we're going to put the virtual machine, so we select **Stellar Platform VMs** and choose **Next**. Okay. Now it's going to ask what sort of networking do we have?

06:00-07:00

Our networking...we can see with that's the networking we previously configured, Stellar management network. That's our port group previously configured that was bound to vSwitch0, which is connected to–I can't remember the name of it Vee NIC0. I'm not sure what that was. The network interface is the physical network interface card that will go out and communicate with our other network and security devices. We have to choose how we're going to provision this virtual machine on the disk. What we're choosing between is thin and thick. We choose thick. What this says is that we've told the virtual machine it's going to have 500 gig of disk space and it provisions all of that 500 gig from creation of the machine rather than if we did thin provisioning, it would create the disk space as required, but we want to have a highly effective and efficient virtual machine, so we will provision it with thick provisioning.

07:00-08:00

We want to uncheck so that, by default, when you get to this screen it'll have **Power on automatically** checked. We want to uncheck that because the moment we click **Finish**, it'll go ahead and try and turn on the virtual machine. We have more configuration steps to do, so we don't want it to boot and then have to turn it off to do that configuration. Uncheck **Power on automatically** and select **Next**. We'll be shown a summary screen of what we've chosen. So we can see we've got a product there: Aella Data Processor. We've got a virtual machine name, the one we made up sc-dp-dl-master, the file where is it coming from? That came from our OVA file and which datastores are going to be put in? What sort of provisioning are we going to do? Which network are we going to use? What is the guest OS? If we're all happy there, we select **Finish** to continue.

08:00-09:00

Once we click **Finish** we can see that the beginning of the creation of our Data Lake virtual machine. It's in the process of loading that OVA file and all of the files within that bundle the files to create the virtual machine. You can see there it's uploading disk at the moment. It's 13 percent run. Depending on where you are uploading this file from will take a few minutes to maybe ten minutes depending on the speed of whether you're doing it locally or or over an internet connection. Okay. When that's complete, you'll see the next slide. We can see in the next slide we have a created Data Lake. We have a Creative Data Lake virtual machine has been created. But there's a few more steps to do at the moment. If you look, CPUs 16, well

that's not enough for a production network. Memory, that's not enough for a production network. There are a couple of edits we need to do before we continue.

09:00-10:00

First one: We're going to edit the Data Lake by choosing **Edit** from the menu. We're going to enter the correct CPUs. We're going to add 40 to this selection because we need 40 for a Data Lake virtual machine, and then select **Save**. We need to edit the Data Lake virtual machine and enter the correct RAM, so that correct is 136 gig. In this example with the checkbox reserve or guest memory, that just means I'm not going to share it if I'm not using it, it's not available to be shared with any other virtual machine. Now on this ESXi host we only have two virtual machines running. We'll have the Data Lake Master and we'll have the Data Analyzer master. There's plenty of memory for each of them. There is no problem with clicking Reserved here. But we should make sure whatever the device is the host is that are running these virtual machines that the memory allocated is enough otherwise it will run into problems if that memory can't be reserved.

10:00-10:50

We click Reserve and then select **Save**. Alright. To create the Data Lake virtual machine we're going to validate it one more time. This is like when we were creating the datastores. It's giving us a summary of what we're creating so we look at the CPU, the memory, the disk, we can see here in this example, we've added the summary of the 12 terabytes. We'll show you how to do that in a later lesson and it's ready to be turned on.

The lesson, *Virtual Machine Creation Data Lake*, is complete. You should now be familiar with creating the Data Lake virtual machine. Thank you.

L03B-DEMO: Data Lake VM Creation (08m:10s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Virtual Machine Creation - Data Lake. Virtual Machine Creation - Data Lake*. This lesson demonstrates the creation of a Data Lake virtual machine in ESXi. The demonstration includes downloading the Stellar Cyber software, creating the virtual machine via an OVA file, and then changing and checking the CPU memory, network adapter, and hard disk requirements for the virtual machine.

Let's start with the software download. If we go to the documentation we can see one of the first things to do under the before you begin, we can see step three here is download the Stellar Cyber software.

01:00-02:00

To do that we need credentials, and you can get those from Support as it says here contact Stellar Cyber at support at stellascyber.ai for login credentials and a one-time password. Or the license key or an OTP will also see it called. Let's start with this. We'll click on that. It's going to ask us for a username and password that has been given to us from Support. The username and password we enter and then sign in, and we can see here, we are downloading some software one download in progress. Again, if we click on the down arrow we can see our download is that software is coming down. Okay. We can see that that software has downloaded. We're ready to go to our install.

02:00-03:00

The first thing to do is go back to ESXi. Here's our host with zero virtual machines. The first thing to do is select **Create / Register** a virtual machine. We're going to deploy this not by creating a new virtual machine but deploy a virtual machine from an OVF or an OVA file. You can see that the file we downloaded was an OVA file. We select deploy via OVA file. Select **Next**. We have to give our virtual machine a name. Let's call it Stellar Cyber Data Lake. I'm going to call it master because at some stage in the future we are going to upgrade our deployment to cluster deployment that may have additional resources. Let's say that this is our Data Lake master. You can either drag and drop a file here or click to select.

03:00-04:00

We're going to click to select. We're going to browse for that file. Here is our file Data Processor that we just downloaded, and open. Now we have the name of our virtual machine. We have an OVA file that's going to create the virtual machine and we select **Next**. Okay. It's asking where are we going to store our virtual machine? Well, before we created some data stores, and we said we would create our Data Lake our Data Analyzer virtual machines in a datastore called Stellar Cyber VM, so we'll select that. There is plenty of room. Remember: 500 gigs for each of those virtual machines. We've got a capacity of 1.75. Tiny bits being used which will be just overhead in the configuration of that, so plenty of space. Select **Next**. Alright. Now it wants to know what is the virtual machine network mapping we're going to use.

04:00-05:00

Well, that's the latest when we created it. If we look there is the VM network, but we're going to go with the port group that was created in a previous lesson called *Stellar Cyber Management*. We're going to do thick provisioning, which means it expands the disk on creation rather than creating disk as it's required. We are definitely not going to power on automatically because as soon as this is being created we're going to want to do some edits before we power it on. So with those selections made, select **Next**. Here is a summary of our machine if everything's okay and there's actually not a whole lot to configure so far

so that summary looks good to me. We click **Finish**. The moment we do that, what we'll see down the bottom here is the creation of our virtual machine. We can see it up here: Our Stellar Cyber Data Lake master.

05:00-06:00

Down the bottom you can see in the recent tasks is that disk has been uploaded from the file on this local machine to ESXi. It's creating our file as we speak. Our disk and our virtual machine so you can see that it's uploading right now, it's nearly finished. As soon as that's finished we'll go ahead and do some edits. Three green ticks so we're ready to continue editing our virtual machine. Now that our virtual machine has been created we need to do some edits. To do that we'll select the virtual machine. Here we are. We can see the defaults. We can see 16 CPUs. That's not enough. We can see memory 64 gig. That's not enough. 500 gigs that's fine. Let's go ahead and edit this and make sure that we configure it correctly. We select **Edit**. Alright. If we go back and remember from our documentation let's go back.

06:00-07:00

Our documentation if we look at the requirements, we had the Data Lake is what we're configuring. We need 40 CPUs. We need 136 gigs of RAM and 500 gigs of disk. If we go down the bottom here, because we're doing this in a test environment or a training environment, the minimum you could have is 16 CPUs, 128 gig of RAM and 500 gig. We're going to go with the big deployment. Let's go with forty one thirty six and five hundred. CPUs we want 40. Scroll down 40, 136 gig, 136 gig, 500 gig of disk, and our network adapter is correct. Right there we could click **Save**, but one of the things we do actually want to do is fix our memory.

07:00-08:10

We go down here and we say we want to reserve all of our guest memory. I did that by expanding this little triangle next to the memory and then chose reserve memory. That means that it won't share. It's always available for our Data Lake virtual machine. Finish that. Click **Save**. Alright. There is our Stellar Cyber Data Lake Master with 40 virtual CPUs, 136 gig of RAM, 500 gig of hard disk space, and the correct network adapter of Stellar Cyber management. We can see that that's connected, as in it's connected to our virtual switch, which is connected to that physical NIC, which is how we'll get to it, to manage the device. We are ready to power on.

The lesson, *Virtual Machine Creation Data Lake*, is complete. You should now be familiar with creating the Data Lake virtual machine. Thank you.

L04 - DATA LAKE VIRTUAL MACHINE CONFIGURATION

L04A-PRES: Data Lake Configuration (16m:05s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform Data Lake Configuration. Stellar Cyber Platform Data Lake Configuration.*

Okay. We've configured our Data Lake virtual machine in ESXi. We're just going to start by looking at the summary of that before we power it on and validate it one more time. You might want to check the CPU, check the memory, check the hard disks have been configured. You can see in this example, there is only one network adapter, and there is only one hard disk. Now for a Data Lake we'll ultimately expect two hard drives, but for this one we just have the one. Let's power it on. We select the virtual machine, and then choose **Power on**.

01:00-02:00
When you do that just wait a few moments and you will be able to click on the virtual console. That black square that's sitting under console and monitor in this image you can click that and it'll pop open to a window, and that is the console to our virtual machine host. Until we do anything with our IP networking we couldn't SSH to this device from a networking device, we have to do it from this console. Power on the device, wait a couple of moments, and then click on this console. We can watch the machine boot the Data Lake boot the Data Lake virtual machine boot. After it has booted you will be presented with the Data Processor login prompt. There are credentials for that. These are default credentials: `Aella` (a-e-l-l-a) is the login name and the default password is `change me` (c-h-a-n-g-e-m-e).

02:00-03:00
We will enter `Aella`, and then the password `changeme`. You can see immediately you are required to change your password immediately root enforcement. We've got to go enter it again. The first password to enter is the existing password, that is `change me`, and then followed by your new password and then confirming your new password. Once you've done that, then we'll be left at a prompt ready to configure this Data Lake virtual machine. The first thing we need to do is to give it IP addressing. Now we'll use DHCP by default so we can use the DHCP if there's that on your local network, but if you want to give it some static IP addressing, which is what we're going to do here, we'll go ahead and enter these commands. We can type in a `set interface management IP`, and then give it an address.

03:00-04:00
In this example our address is 192.168.1.175, and then we've got to give that IP address a mask. In this example it's going to be a /24 or a 255.255.255.0 network mask. Set interface management IP 192.168.1.175/24. When you press Enter, you can see there after setting interface IP do reboot and reset to apply the configuration. Yes, we will do that, but we won't do it just yet. We'll just put a couple more IP addressing in: one for the gateway and one for the DNS. You can see those two commands set interface management gateway, and then give it an IP address In this example it is one nine two

one six eight one dot two five four for a gateway and then a set interface management DNS. We'll give it a DNS of 8.8.8.8 for DNS.

04:00-05:00

Now the first of the show commands we've seen if we do a `show interface`, we can see we have one interface, the management interface and the link is up because it's connected to our port group inside our VM machine, which is connected to our vSwitch0, which is connected to that physical NIC that's connected to an external switch. We have a link of up And an IP address we can see there of that 192.168.1.175 with our mask written with a 255.255.255.0 or a /24. It's all good there. Everything looks fine. Once we've done that, we can SSH to the box and you can see here from Mac in this example it's SSH, and then the username Aella (a-e-l-l-a) @ and the address we just gave at the 192.168.1.175. We will have to confirm our fingerprint with a **yes**.

05:00-06:00

You can see at the bottom of this page, we have that same prompt that we had in our console. Now the reason for using SSH over that virtual console is we need to have a bit more control of what we can do in an SSH host depending on what client you're using, the console in ESXi is not very forgiving. For example, if you typed in `show ?`, it will run up the screen and there's no way to scroll up. Where if we had a good SSH client, we can scroll up and see more of what's available in our show's question mark. There you go. Log in with Aella, use the new password that you created in that last section during the boot. Alright. Just to confirm, `show interface` There it is again: 192.168.1.175 with a /24 bit mask and `show gateway` gives us the information we want: 192.168.1.254.

06:00-07:00

All good there for our interface and our gateway. The last one to check is our DNS configuration. We do a `show DNS` to show 8.8.8.8, which is what we configured. So the test there with a shell dig plus short meaning it's just shortening the output of what's been returned of www.Google.com and we can see a bunch of responses there. We actually know that the configuration is working correctly. I show a host name. We'll just `show Data Processor`. We actually want to give it a host name. When we go into the UI we will see this host name listed when we look at the Data Lake. `Set hostname sc-dp-dl-master` is code for Stellar Cyber, DP, Data Lake. Sorry. Stellar Cyber, Data Processor, Data Lake, master. It's any host name you'd like to give it.

07:00-08:00

Again, it's asking you to reboot and then reset. At this stage that's what we're doing. From within our SSH client we will issue the `reboot` command and you can see it's closed that session. The device, the virtual machine will now be rebooting. If we go back to the console, because it's the only place we can watch it, we can't SSH to a device that's booting, but we can watch from the virtual console within ESXi we can watch it boot. That screenshot is just an example of some of the messaging while it's booting up. We've received our login prompt and we can SSH back into our device with the username and the host IP address of that 192.168.1.175. There we are.

08:00-09:00

With our new password, not changeme, the new password. Confirming that we are back into our Data Lake. We've booted the box, we've changed the IP addressing, and now we're ready to configure a little further. Now we need to set it a role. Our device, our virtual machine is a Data Lake, but it's not really known as a Data Lake at the moment. All it was was a host with a host name. The set role command and you can see the available roles at the top. AIO is for an all in one. We can create a Data Processor that's got a Data Processor and a Data Lake all in one, not for production networks. These are for test networks. Makes the deployment a little bit easier. Resource is we'll see resource used when we create the Data Analyzer because you can see in that in that list of availability there's no DA, no Data Analyzers so we create it as a resource first and then in the UI create the resource into a Data Analyzer.

09:00-10:00

You can see there what we want to choose is DL-master. A set role DL-master, and we can see there that now in the screenshot below it's pretty important. When you do this if you do it too quickly after booting, you can see in that screenshot of the set role DL-master, it comes back and says CLI service is still initializing. Please wait and try again. Just wait a few moments and you can actually press the up arrow and then enter and it'll issue the same command again. It's just waiting for that CLI service to boot. Right. Now we have this as part of a cluster. It's not an all in one, so we need to give it a cluster name. A set cluster_name, and then name the cluster. In this example, the cluster's going to be called sc-dl-cluster. This is the Data Lakes cluster that we're configuring here and a show cluster_name...

10:00-11:00

...will show that that command was successful. Well, the Set Cluster Name Success was the giveaway that it was successful, but just to test a show cluster underscore name will give us the name of that cluster. We can see there sc-dl-cluster. We have to give the cluster a size. The command for that is set cluster_size and it's a cluster size of one. When we do a show cluster size just to confirm, again we're just confirming if we looked at the output from the set cluster size one, we can see set cluster size success, but just to confirm show cluster_size and we can see the output of that command is one. At the start you can see there set the size of the Data Lake cluster. Well, at the start of this there's only one. There is a Data Lake master and then we'll increase it by one each time we add a worker node. We'll use a set cluster_size and a show cluster_size to see that.

11:00-12:00

A show interface. There is no difference here. We're still just confirming our 192.168.1.175. The reason that's important is we're going to do a set cm, which is the configuration manager. A set cm for this deployment will be using the IP address of the Data Lake Master's Management IP. A show interface gives us that output of 192.168.1.175. A set cm of 192.168.1.175 is the command we need. To confirm that, we do a show cm we can see there the output 192.168.1.175 has worked. We have set the configuration manager correctly. Right. The final thing to do is to set an OTP because after we've configured this...

12:00-13:00

...this virtual machine will go and pull images from Stellar Cyber repository of data to create this virtual machine. Now without a licensed OTP or a one-time password, our license without a valid one of those, it won't pull down all of the images. We need to do a `set otp` and then type in our OTP that we received via email from Stellar Cyber as part of the deployment. That example there–a123b432c123–is a made up one. It's just an example, so please don't type that in. It will not work for you. Once we do a `set OTP` and then we can do a `show otp` and we can see that it's been verified successfully. Again, the a123 is just a mock number so please don't use that. It will not work.

13:00-14:00

Right. Now we have typed in We've entered the configuration for the IP addressing. We've set up the clustering. We've given it a host name. We have rebooted the box and now it's time to do a reset. With the configuration commands of the Data Lake in the CLI complete, it's time for a rese. We issue a command `reset` to confirm that because it's going to go and reset all of our pods or all the services running and start pulling down new images. It asks us for a confirmation with a capital YES. So it says, "Are you sure? Type upper case YES," and then off it goes. It will start resetting the device. To see where it is up to, we use the `show status` command. What we're looking for here is you can see how many pods are running and it tells you how many it's expecting to see.

14:00-15:00

In this case at the very beginning there are zero pods running, and it's expecting at least 39 to be running. The image is installed on host. Depending on what type of deployment that is and what images are required, you will see that 33. That number will get up well into the eighties and sometimes I've seen into the early nineties. There are images that have been pulled down from a Stellar Cyber server to pull those images down and then to get those pods up and running. It can take five, ten, fifteen minutes depending on loads and depending on your network activity. Continue to do a `show status`. We've got a couple more screenshots that will show us that. Here is another `show status` and we can see now we have some pods running. In this output we can see we have 17 pods running, still not even halfway of how many we're expecting to see.

15:00-16:05

There are 59 images, still not near the high eighties or early nineties we're expecting, so continue to be patient while those images come down and those pods start running. Let's look at another one. We are finished! All pods are running. 90 images are installed on this host, and the big green "system ready," that's what we're looking for. That's great news. Everything's come down. Now clearly our OTP is working, and you can see there the OTP was verified. Again, that's just a mock OTP address but you'd see your real one there a system ready. That's it. That's the Data Lake configured up until this stage from a CLI perspective. The next thing we'll need to do is to jump into the user interface, the web pages of our Stellar Cyber platform, and do some more configuration in there.

The lesson, *Stellar Cyber Platform Data Lake Configuration*, is complete. Thank you.

L04B-DEMO: Data Lake Configuration (17m:10s)

00:00-01:00

The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform Data Lake Configuration. Stellar Cyber Platform Data Lake Configuration*. Demonstration of the initial console CLI configuration for the Data Lake.

In this demonstration I will show you the configuration steps of the Data Lake console including the initial login to the console requiring a password change, the networking configuration settings and validation, setting hosts and cluster names, role and configuration manager settings, and of course then the resetting and watching it come up all ready. Let's start with powering on the Data Lake virtual machine. Alright. Here is our ESXi host.

01:00-02:00
We'll go check our virtual machine. There it is, our Stellar Cyber Data Lake master. We can either check the box here and choose **Power on**, or you can select the virtual machine and click **Power on** from here. Let's do it from here because I want us then to open up this console to watch the machine boot. Let's do that power on. We can see down here the recent task is Power On virtual machine, successfully completed. We click on this. We'll get a console. Let's now watch the Data Lake boot. Just about finished. The change in font is a bit of a giveaway. Here we are. We have a login prompt. Alright. The username is `aella`, a-e-l-l-a.

02:00-03:00
The first password, the default password is `changeme`. All lowercase and no spaces, c-h-a-n-g-e-m-e. We're then asked to immediately change the password. We entered `changeme` again. That's the current password, `changeme`. I'm going to enter a new password and confirm the new password. Here we are in the DP. Alright. The first thing to do, let's set some IP addressing because if we do a `show interface` we can see we have an IP address, but this is a DHCP IP address and I don't want that. I actually want to set an IP address. Let's start with `set interface` tab, space `m-a-n`, tab, IP...

03:00-04:00
...and the address we're going to give it is 192.168.1.234 and we'll give it a mask a `/24` bit mask **Enter**. Now it does say here to reboot and reset, which we will do. We're just going to put a couple more configuration settings in. We could do a `show interface` to see that our new IP address of the .234 has been set. We're now going to give it a gateway and some DNS information. `Set interface` tab, `m-a-n` tab to complete that command, `gateway` tab, the gateway IP address is `192.168.1.254`. **Enter**. We could do a `show gateway` command to show that the gateway configuration has been set. We'll do a `set interface`...

04:00-05:00

...Management dns to a 8.8.8.8. Enter. Show DNS. Okay. We've got all those now. Okay. Now that we've got a configuration on there I want to SSH the box so that we can have a better console rather than this virtual console within ESXi so I'm going to close that down and open up a SSH. Okay. Here's my PuTTY, which I'm going to use for SSH. I'm going to load up a profile that I've configured earlier, and (select) Open. Okay. The first time I'm going to use it here we need to accept the fingerprint, so accept. So aella to log in and our new password, not changeme, but our new one.

05:00-06:00

Here we are. Have we got the right device show interface? Yes we do. We can see there that we've got 192.168.1.234. Show gateway. Yes we've got gateway, show dns. Yes. We've got the correct DNS. The first tests we can do here is let's ping. We need to do an !ping. Let's ping our gateway 192.168.1.254. Great. We can do that. Let's ping our DNS. !ping 8.8.8.8. This will check that our gateway is working. We can ping our disk. The last one to check that our DNS settings are correct. Let's do a shell dig to google.com, and we can see that is also working because we've got an answer there of that 142.251.46.206.

06:00-07:00

Okay. They are all set correctly. The next thing is to set the hostname, reboot it, log in, and then set some role information, so standby why we do this. We do a show hostname. It's just what we've got. Let's set hostname to be sc-dp- (for Data Processor) dl (for Data Lake), and it's going to be the master. **Enter**. Okay. You can see there it says to reboot and reset, which we will do. We will do that. Well, let's do that right now. We've lost our party session. We'll close that. Let's go and look at our ESXi console to watch it boot.

07:00-08:00

We'll see as soon as this font changes. There we go. Now we can log in. We'll be able to open up our party session again. Let me do that. aella, the new password, and here we are. Okay. A show hostname. We can see that we've now got our new host name. Fantastic. Alright. The next thing to do is to set the role. We're going to do a set role. If I enter we should get a list of what's available. We're going to set ALs up as a DLmaster. Set role DL-master. Okay. We have it now set as a DL master. The next thing we need to do because this is not an all in one is we need to set the clusters up. We're going to give this a cluster name. That gives us the ability to have multiple Data Lake nodes run into that. They run in a cluster. Let's set that up with a set cluster_name tab. Yep. Let's say cluster name. We're going to give this a Stellar Cyber dash Data Lake cluster (sc-dl-cluster). That's our name. Set successful, and we're going to do a show cluster_name. See that was set correctly. Now we need to give the cluster size. Set cluster this time at size not

name (`set cluster_size`). We're going to give cluster size and one because at this stage we only have one node being this DL master running in this cluster. It's that size of one. We can do a `show cluster_size` to confirm that worked. There we are at one.

09:00-10:00

Okay. The next thing we need to do is set the cm, the configuration manager. To do that we're going to set it to the same IP address as our Data Lake master. If we do a `show interface` we can see that's our address, our 192.168.1.234. We're going to do a `set cm` to `192.168.1.234`. Fantastic. A `show cm`. Fantastic. That's been set correctly. Now we need to set our license because after this it's going to go pull packages down from the Stellar Cyber site. Now without a licensed version of this, it won't pull all of the packages down. Now we do that with a `set OTP` command. You can get that in your welcome pack that you'll receive or from Customer Support. Don't use this one. We're going to cover this up. This is our training OTP, so it won't be useful to you...

10:00-11:00

...but let me go ahead and set that command now. Okay. There's our `set otp`. You can then do a `show otp`, and it will show you what you've got that has been successfully verified. The last thing now to do is to do a reset and watch the packages come down. Let me do that now. We do a `reset`. It says caution. Yes. We do want to do a reset. We have to answer with as it says down the bottom type uppercase `YES`, so Y-E-S in uppercase. **Enter**. Now it's going to go ahead and start pulling down images and starting to run the pods. The way we can verify what's going on with this when we get a command prompt back is to do a `show status`. I'll walk you through the process of how to do that. It takes a little bit of time while all of the packages or the images are pulled down.

11:00-12:00

If we do a `show status`. Okay. We can see there what we're looking at or want to highlight your attention to is zero pods running and it has an expectation of at least 39. We can see that there are 27 images installed on the host. That number will increase. We can see there in red there are microservice images missing. They are being pulled down right now. A lot of red there that's going on, so I will continue to run a `show status`. What we want to see is that the pods running is increasing and the images installed is increasing. I can just press up arrow for `show status` again. **Enter**. Alright. We still have zero pods running, but we now have 30 images installed. We can see that the images are coming down. Okay.

12:00-13:00

We could see up here it was 27 images, and now it's 30 images, but still zero pods. We'll try that again. Up arrow `show status`. What have we got now? Thirty-two, so still no pods running. We'll keep an eye on that and 32 images installed on the host. I will stop talking.

There's not much for me to say, but I will continue to press up arrow to bring back the command `show status` or you could just type it in. But we'll speed up this part of the video so we don't have to wait and watch for the entire pods to be running. I'll stop every once in a while and chime in of what I'm seeing and what I'd like to draw your attention to until we have it complete. Stand by. Alright. There is something to mention here. We can see in the last version of `show status` that we now have six pods running, so that has started to increase.

13:00-14:00

We can also see from the last version of this that the 34 images installed on the host has increased to 37 images. I believe it gets in excess of 80 images. It depends on what sort of deployment it is and whether it's a DA or a DL, but it gets I think maybe even into the 90s for images. They're coming down right now as we're waiting and watching, but we do want to see the pods running increase and the images installed on the host to also increase. Let me just check one more time while I'm talking to you. `Show status.` We were six and 37. We're now eight and 40. I will pause while we wait for these to come down and spin up. Okay. What can we see here? We are up to 26 of at least 39 of the pods running and 66 images installed.

14:00-15:00

It is progressing as planned. I do keep checking it occasionally. I've witnessed that these stall. If that's the case, you just do another `reset` and it goes as a clean up and pulls down. It actually doesn't remove what it's already brought down, but it does reset all the pods. If it does stall, that's something that you can do is just to do another `reset`. But do keep an eye on it and make sure that the images are coming down and the number of pods running is increasing. Let's do another `show status`. Let's see where we're up to. Thirty-six of 39. The number of pods is very close. Eighty-six images on the host. That is also very close, so not too long now and we will have our Data Lake up and running.

15:00-16:00

Now although I didn't mean to do a `show interface` before, it's probably quite telling for us to see those RX bytes is at 19 gig that shows all these packages that are coming down all these images as it says here these 86 images installed on host, they'll come in through this management interface in the RX byte. I'm going to actually do that again. Let's do a `show interface`. We can see that it is growing. Just in the last time I did that it's gone from 19.6 to 20.3. Let's do a `show status` and see where we're up to. Okay. Thirty-six and 89. We have three more images and still the same amount of pods running. We're getting very close, but we're not there yet. Standby. What we can see now is 89 images have been installed on host. All images and below that you can see all images to run microservices are ready.

16:00-17:10
All images to support platform are ready. While we're waiting on now is those 36 pods running to be as it says at least 39. Standby. We're very close now. Okay. We can see that we now have all 89 images are installed in the host. All pods are running. That's been the change from 36 to 37 to 38. Now we have all pods are running. Just the last couple of things that it's cleaning up around the manifest file. We're nearly there. As you can see now all pods are running. All images are installed on the host and the most important is what's there in green writing System Ready. We are ready to sign into the UI.

The lesson, *Stellar Cyber Platform Data Lake Configuration*, is complete. Thank you.

L05 - DATA ANALYZER VIRTUAL MACHINE CREATION

L05A-PRES: Data Analyzer VM Creation (08m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Virtual Machine Creation - Data Analyzer. Virtual Machine Creation - Data Analyzer*. Just looking at the minimum requirements for the DA the Data Analyzer we can see in the circled area, the Data Analyzer requires 44 CPU or vCPU, virtual CPUs, 64 gig of RAM so a little less than the Data Lake and the same 500 gig for the actual virtual machine itself. That's on the ESXi device. To go find those minimum requirements, please go to the documentation and you will find this is just a screenshot of what's available in those documentations in the Knowledge Base.

01:00-02:00
The minimum requirements for a demo environment or training environment, you can see we've dropped the DA CPU from 44 down to 16. RAM remains the same 64 gig and of course the disk space in the virtual machine in ESXi remains at 500 gig. We will have plenty of space for that. You can see the DA and the DL virtual machines will be stored in the same datastore. If there is sufficient room on our datastore, you can put the Data Lake there as well for a demo or for a training environment. In a production environment, Data Lake needs to be in its own datastore for performance. To create this Data Analyzer, this virtual machine, we'll navigate to virtual machines on the left side and then select the **Create / Register VM** to create it. You can see we've already created the SC DP DL-Master.

02:00-03:00
Now we're going to create the SC DP DA-master. Navigate to **Virtual Machines**, click on **Create / Register VM**. We're going to use the same file that we downloaded and used for the Data Lake, the same file to create the Data Analyzer. We'll use the OVA. Select the middle one: **Deploy a virtual machine from an OVA or an OVA file**. We'll be using an OVA file, so it's that middle selection, and then click **Next**. Enter the name for your DA virtual machine in this example. Of course you can name it whatever you like. Our example is SC DP DA (for Data Analyzer) dash master (SC DP DA-Master).

This will be the first one in a scalable environment. We're only going to create one, but if we were to have many this will be the master virtual machine. We can of course have worker nodes...

03:00-04:00

...DA worker nodes to add to this to increase the scalability of this Data Processor. We give it a name, and then we either drag the file into that open space, it starts off as being blue open space, or if you click in that space, it'll pop up a menu where you can browse to find that file. We're looking for that aella Data Processor 4.3.7. Remember, we're doing 4.3.7 so in a later lesson we can upgrade it to 5.1.1 and show you how that process works. When you've done that, select **Next**. Okay. Now where is it going to be stored? This is choosing the datastore of where your virtual machine will be placed. We've created two datastores: the SC Data Lake storage, that's where we're going to put our Data Lake, but where we're going to put our actual virtual machines is in the Stellar platform VM. We select **Stellar Platform VMs** and choose **Next**.

04:00-05:00

Okay. Networking, we will go choose a network connection for the virtual machine the DA at this in this deployment in our basic deployment we'll only have one network interface. We're selecting the Stellar management network in this example. For disk provisioning, we choose thick, which makes the space available at the creation of the virtual machine, and then we uncheck our **Power on automatically** because there's some changes there's some edits we want to make prior to turning on the machine. Make sure you uncheck that if you forget and you turn and we've clicked **Finish**, all it's going to happen is it's going to boot straight away and we'll have to power it down so we can do these edits before turning it on again. Easy here just to uncheck the default because it is checked by default. Uncheck the **Power on automatically** so that we can make these edits. When you've done those changes, select **Next**.

05:00-06:00

You'll now see a window that shows you your summary of your virtual machine that's about to be created. You can see the product is aella Data Processor. It is a VM. That's the name we gave it. Make sure that the datastore is correct and your provisioning is thick as we said, and the right network configuration. If it's not, you click the back button and go and fix that, else if it's all ready to go, only thing left to do is select **Finish**. Now once you click **Finish** you can see down the bottom of your ESXi webpage, you'll see that the progress and at the moment it is uploading that OVA file from wherever it was that it was stored to your virtual machine to that datastore. That's where it's been created. Just wait a few moments depending on where it comes from, if it's a slow network connection it might take a little bit longer, but it takes a couple of minutes and it will upload that virtual machine.

06:00-07:00

We can see now that the virtual machine has been created. There it is. You can see that little green dot for the SC DP Data Lake or DL master. That means it's on and running our SC DP DA master. It does not have the little green icon. It has not been turned on yet, but it's ready to turn on except we're going to make a couple more edits prior to turning it on. We're going to select the SC DP DA Master and make some more edits. Once you've selected it, this is the screen you'll be presented with. At the top there on that menu bar there is an Edit button so we select the **Edit** button. You can see here

we're going to change the CPUs. This will be a production environment example so a minimum of 44 CPU, virtual CPUs. You can see there is a reservation of those CPUs. Now this is up to you of what your deployment looks like.

07:00-08:00
In our example here, there is one ESXi box that's hosting only the two VMs, one VM for the Data Lake master and one VM for the Data Analyzer master. In that scenario, it's not necessary to reserve the CPU as they'll be shared between the two Stellar Cyber virtual machines. But if you have a larger environment where your virtual machines, your DA DL Stellar Cyber virtual machines, are on a shared environment with other servers, other virtual machines, then it would be wise to reserve the CPU so they're always available to these virtual machines. You would check that reservation and then click **Save**. Okay. Another edit for the virtual machine is to reserve the memory. Again same thing as the CPU.

08:00-08:50
I would defer to say yes, reserve it. It's probably a good habit to get into to make sure that when these devices spin up that they've got their CPU and memory available to do all of the functions that are required. When you finish that, click **Save**. At the end of it we'll be given back our summary and you can see there that the CPU is correct, check that the memory is right, the hard disk is correct, and the network adapter is right. Once that's all confirmed, your virtual machine is ready to be powered on.

The lesson, *Virtual Machine Creation Data Analyzer*, is complete. You should now be familiar with creating the Data Analyzer virtual machine in ESXi. Thank you.

L05B-DEMO: Data Analyzer VM Creation (08m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module...is *Virtual Machine Creation - Data Analyzer. Virtual Machine Creation - Data Analyzer*. Demonstration of the creation of the Data Analyzer virtual machine in ESXi.

In this demonstration, I will show you the creation of the Data Analyzer virtual machine in ESXi, including the system requirements. We'll have a look at the documentation again. We've already downloaded the Stellar Cyber software when creating the Data Lake, so we don't need to download that again. We'll go through creating the VM via the OVA file that we just talked about downloading, and we will change and check the CPU memory and network adapters for the virtual machine.

01:00-02:00
Let's get started with the requirements of the Data Analyzer virtual machine. Here we are in our Knowledge Base. Just to navigate there rather than searching we could search or navigate. Let's navigate. We're **INSTALLING | Installing a Data Processor | Installing a Data Processor in VMware as a cluster**. You could do the same thing. Let's do it. Let's do it. If I go back to home, if I type in `VMware cluster` in the search, let's see what shows up. There we are, I*nstalling a Data Processor in*

a VMware Cluster. Just so you know, up at the top it gives you a UI: Installing | Installing Data Processor | Installing a Data Processor in VMware (Cluster). That's the same way we've been navigating. Either one works. It will then also highlight. You can see any of the words that we had put into the search. I'm just going to remove that from the URL...

02:00-03:00

...so we don't have those words. Okay. Before we begin some system requirements. We're talking about the DA this time, the Data Analyzer. So over here in this column we can see the host needs 44 88 cores because we're going to have 44 for the Data Analyzer, RAM 64 gig, OS for the VM 500 gig that's what we're going to make sure of. Then down below from a minimum perspective, and we could lean towards the minimum being that this is a demonstration / training deployment, 16 CPU rather than the 44, we still need the 64 gig of RAM, and we still need the 500 gig of space. Let's go ahead and install that. Okay So here we are in our ESXi host. If we select Virtual Machines, We can see that we have a Stellar Cyber Data Lake master but we're going to create a Stellar Cyber Data Analyzer master. Select **Create**.

03:00-04:00

We're going to do it from an OVA file. Choose the middle selection and **Next**. Give it a name: Stellar Cyber Data Analyzer Master. We need to select that OVA file. I'm going to browse by clicking there. I'm going to select the Data Processor OVA file. Select **Open**. Now we have that ready to go. Select **Next**. Where would we like to store this? We're going to store it in the same place as we stored the Data Lake Master, that is in the datastore Stellar Cyber virtual machines or VMs. Select **Next**. Alright. Which VM Network are we going to use–that port group? We can see we've got two to choose from. We're choosing the one we created earlier Stellar Cyber management, and we're going to thick provision, which means it provisions the disk space rather than expanding as required.

04:00-05:00

We are not going to turn it on automatically because we want to edit before we go further. Select **Next**. Alright. Here's a summary. In case you've made some mistakes, you can always click back to go and fix those. We have an aella Data Processor. The name of it is Stellar Cyber Data Analyzer master. You can see it's pulled this information, which has come from that OVA file. The datastore is on the Stellar Cyber VM. It's thick provisioning. It's going to use the Stellar Cyber management network for networking. Its guest OS is in Ubuntu Linux. That's it. Let's click **Finish**. Down at the bottom under **Recent Tasks** we'll see that it's created the VM and it's starting to upload those disks. Let's let that complete. Okay. Nearly there.

05:00-06:05

We now have our Stellar Cyber Data Analyzer master. We've got to make some edits to this before we go and make sure we've configured it correctly. Let's go in here and we've selected it. We're going to choose **Edit** to edit the virtual machine. We were going to give it 44 cores. Let's give it 44. Sixty-four was the memory, 500 was the disk. It's got the correct network adapter. We click **Save**. All done. All there's left to do now is power it on.

The lesson, *Virtual Machine Creation Data Analyzer*, is complete. You should now be familiar with creating the Data Analyzer virtual machine in ESXi. Thank you.

L06 - DATA ANALYZER VIRTUAL MACHINE CONFIGURATION

L06A-PRES: Data Analyzer Configuration (08m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform - Data Analyzer Configuration. Stellar Cyber Platform - Data Analyzer Configuration.*

Now that our virtual machine has been created in ESXi, it's ready to power on. Just one last time before you power it on, check that you've got the CPU correct, the memory correct, the data storage is the 500 gig on the right datastore. Networking is correct. If all that's good to go, we're ready to power on. After you've selected the VM, the virtual machine, from this **Virtual Machines** tab on the left, then choose the **Power on** button at the top. Now while it's powering on you can see that in the virtual console.

01:00-02:00
We're unable to SSH to this device remotely because of course it doesn't have any IP addressing, so we can only use the virtual console to see what's going on. At that stage, once it's complete in the boot, we'll be presented with this screen where we log in for the first time. The first time we log in at the login prompt we enter the following default credentials: `aella` (a-e-l-l-a) is the default login name, and then the default password is `changeme` (c-h-a-n-g-m-e). Type that in `aella` and `changeme`. The very first thing you're required to do, as it says on the screen, you're immediately asked to change that default password. The steps are you enter the default password again. That's the current password of `changeme` (c-h-a-n-g-e-m-e) all lowercase. Then enter your new password and confirm it. Once that's done you'll be presented with your prompt in your CLI, ready to make the configuration changes.

02:00-03:00
The configuration changes: We have got to give it some IP addressing. It will use DHCP by default, so the actual host will receive an IP address if you have a DHCP server on that network segment. We're going to use static IP addressing. The commands required for that is a `set interface management IP`, and then the IP address and the netmask. In this example we can use a /24 for a 24-bit mask. You can see in the output there it translates our /24 into a 255.255.255.0 in the octet notation. We've done a `set interface management IP 192.168.1.150`. As we saw before, our Data Lake is .175, so they're on the same network, the /24 network. The next thing to do is give it a gateway.

03:00-04:00

A `set interface management gateway` is the command. In our example, 192.168.1.254 is the gateway. The final command here for our IP addressing is a `set interface management dns`, and then the DNS address In this example, it is 8.8.8.8. Then a `show command` or `show interface`. `Show int` is enough. If you actually type in `show int` and press tab, it will extrapolate and expand that syntax out so it will actually end up being `show interface`. We can see in the `show interface` command that our IP addressing is correct of the 192.168.1.150. You can see in the output straight after we enter that IP address it asked us to do a reboot and a reset. You don't have to do that immediately, but it is required before we continue too far. To add then the gateway address and the DNS settings is just fine to do those three...

04:00-05:00

...IP set interface commands before we continue. Alright. Some show commands, a show interface. We can see that IP address in there a show gateway command will show us the gateway of the .254, which we configured before, and the `show dns` of 8.8.8.8. Now there's a couple of commands you can use to test that these have worked step by step. I mean the first one I would do would be a pin to the gateway. You can see down at the bottom it's a `shell ping 8.8.8.8`. The other way to do it is `!ping` and then the gateway address. In this example 192.168.1.254 to make sure that we can get at least to our gateway that would prove our IP addressing is correct. The next step would be to ping as it says there shell ping 8.8.8.8 to ping something beyond our default gateway.

05:00-06:00

Again, `!ping 8.8.8.8`. Make sure that our default gateway configuration is working. The final one to test whether the DNS configuration is working, is to do a `shell dig www.google.com`. That will show us that our DNS is resolving. They're the three commands we can use, or four commands we can use to make sure that our IP addressing is correct. Once that's done we can SSH to the device. We know it's now working. We can SSH to aella. That is our login name, at 192.168.1.150. First of all you'll have to confirm the fingerprint with a yes and then you're presented with the login prompt. Now the login name is aella of course, but the password is no longer changeme; it's whatever you've changed it to during that very first login where you're forced to make a password change. Right. `Show host name`.

06:00-07:00

We do a `show host name Data Processor` not terribly helpful to us when we get into the UI and want to identify this virtual machine. We'll do a `set host name` then. The host name we're choosing is `sc-dp-da-master`, or if I'm to expand that would stand for Stellar Cyber Data Processor, Data Analyzer, and it's the master one of those. Again, you see it asks you to do a reboot and a reset. At this stage after entering our IP address in and confirming that it's correct, adding a host name it's now time to reboot. Hit the reboot and you'll see immediately your connection via your SSH client has been closed. We'd have to go back to the ESXi virtual console to watch the device boot up again. We didn't watch that boot up again. We're just assuming it booted up. Here we are. We have SSH back into the box and now we're going to set a role. Previously with the Data Lake we set it a role of Data Lake Master.

07:00-08:00

When we configure a Data Analyzer, we start by just setting its role to a resource. We can see there with the first command `set role resource`. The `show role` command will show that that stuck. We've got the resource as our role. We set the cm. The CM or the configuration manager is the IP, the management IP address of the Data Lake. That's the virtual machine we created four which had the example of 192.168.1.175. `Set cm` to the management IP address of the Data Lake interface, and a `show cm` will show that we've done the correct settings there. Now we need to do a reset. The reset will ask us to confirm with a capital YES, an uppercase Y-E-S to start resetting our pods and pulling down whatever software is required.

08:00-08:25
Now at this stage it's not a DA, it's just a resource. It will pull down the required images that are needed just for a resource. At this stage we can go and watch what's going on.

The lesson, *Stellar Cyber Platform Data Analyzer Configuration*, is complete. Thank you.

L06B-DEMO: Data Analyzer Configuration (09m:15s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform - Data Analyzer Configuration. Stellar Cyber Platform - Data Analyzer Configuration*, Demonstration of the initial console CLI configuration for the Data Analyzer.

In this demonstration, I will show you the configuration steps in the Data Analyzer console including the initial login to the console requiring a password change, the network configuration settings and validation of those settings, setting up the host and cluster names, and setting a role and configuration manager settings, and of course resetting the virtual machine. Let's start with powering on the Data Lake virtual machine. Here we are in ESXi. If we go to our virtual machines we can see we've got two.

01:00-02:00
We've got this one powered on our Stellar Cyber Data Lake Master. That's that little green play icon we can see just above there. We can see that our Stellar Cyber Data Analyzer master is not yet powered on. We're going to select it, and we're going to select **Power on**. Wait just a couple of moments and then we will open up this console to watch it boot. Here we go. There's our prompt. Same as with the configuration of the Data Lake, the Data Analyzer, it's aella. The password, the default password, which will change immediately, is change me c-h-a-n-g-e-m-e. Here we are immediately being asked to change that password. We've got to type in the current password, which is change me c-h-a-n-g-e-m-e.

02:00-03:00
Now it's asking for our new password to enter. Confirm that new password. Fantastic. Here we are in our console of our Data Analyzer master. The first thing to do, set some IP addressing. Okay. We start with `set interface m-a-n` management, IP, and the address with a mask `/24` bit mask for that network and **Enter**. It does say here to reboot and to reset, both of which we will do after we put in

just a few pieces of configuration here. The first one is to set the gateway: set int, tab management tab...

03:00-04:00

...gateway space. The gateway is 192.168.1.254 (`set interface management gateway 192.168.1.254`). Now we've got our gateway. Again, it's asking us to do that reboot and reset, which we will do at the end. We've got to set up some DNS settings, so `set interface`. Look if I just type in two letters and press tab, nothing happens. I need that third one to make it unique. Interface management DNS, 8.8.8.8. Enter. Okay. Fantastic. We've got enough configuration items here that we can SSH to this device now. Let me go ahead and do that. One moment. Okay. Here we are with a PuTTY session or an SSH session to 192.168.1.233.

04:00-05:00

It asks us to accept the fingerprint, accept, and we're going to log in. We're going to log in as aella. Our new password, not `changeme`, but our new password (types). Here we are in our Data Analyzer console. Okay. A couple more things to do now. Let's just check where we are. We'll do a `show interface`. We can see we are on 192.168.1.233. That is the one we're expecting to see. Let's do a `show gateway`. That is correct, .254 is the gateway. A `show dns` will show us the 8.8.8.8, which is correct. Let's test those now. Let's see if we can ping our gateway. So `!ping 192.168.1.254`. We always want to check that the networking is working correctly before we continue.

05:00-06:00

We can ping our gateway is working. Let's ping our DNS server to prove that the routing is working. `!ping 8 .8 .8 .8.` That is working. The final one to test is to do a shell dig to google.com to confirm that our DNS is working (`shell dig google.com`). We can see that's working there with our responsive or the answer of 142.250.189.238. Okay. They're all working now. Now let's give it a host name. What have we got right now? If we do a `show hostname` it is just Data Processor. Let's give it a host name. `Set hostname`. I did a tab then. Stellar Cyber-data processor-Data Analyzer-master (`sc-dp-da-master`). Let's use the same convention we used for the Data Lake.

06:00-07:00

If we do a `show host` let's see what that says. See it's still there. This is where we need to do a reset to confirm that that's going to work. At this stage, let's do a `reboot`, r-e-b-o-o-t. Our SSH session will be discontinued, but we'll go look at the virtual console in ESXi to confirm where it's up to. We'll move that out of the way. Let's have a look. We can see here we are rebooting. Okay. We could log in there, but let's go back to our PuTTY session and run it again. Restart session. Okay. There's our login at the bottom. Type `aella` and our new password. Okay. Here we are.

07:00-08:00

Alright. We are in our Data Processor. If we do a `show hostname` now, that's what we expect to see. That's the name that's going to show up in the UI that `sc-dp-da-master`, That's why we're sort of interested in that because we want to see what it is up there. Okay. Now we need to set the role for this. If we do a `set role`, we can see there that none of those options are for a DA. In this example what we will do is set it up as a resource and then inside the UI, the user interface is where we

configure it as a Data Analyzer. The command for this is `set role`. I could actually just press up arrow here to bring back the last command, `set role resource`. Okay. That looks like we've got it. It's in that prompt now but let's do a `show role`.

08:00-09:15
The role is of a resource. So fantastic. Alright. We need to do a `set cm`. `Set cm` to 192.168.1.234. We want to set the cm to be the management interface of the Data Lake. That's the other IP address. This device as you can see up here is .233. Alright. That is our cm. We'll do a `show cm`. What did I do? What did I do wrong? I typed in the wrong address. I typed in 268, not 168. Of course 268 is outside the range of available addresses. `Show cm`. There. Okay. That's fantastic. We are ready for a reset with a big fat capital `YES`, Y-E-S, to allow that to happen. Okay. That's going to take a while while that's going on. We'll come back.

The lesson, *Stellar Cyber Platform - Data Analyzer Configuration*, is complete. Thank you.

L07 - DATA PROCESSOR CONFIGURATION

L07A-PRES: Data Processor Configuration (09m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform - Data Processor Configuration. Stellar Cyber Platform - Data Processor Configuration.*

This is the first time we're actually going to log in to the user interface of our DP, our Data Processor. Data Processor is made up of a Data Lake and a Data Analyzer. In our example we've got a Data Lake master and a Data Analyzer master, that Data Analyzer master has not yet been configured. It is currently just a VM as a resource. In this module we'll show you how to convert that from a resource to a Data Analyzer. The first thing to do is to use your browser and browse to the Data Lake management IP in our example, that 192.168.1.175.

01:00-02:00
We use admin. It's not aella in the UI. It is admin with a password of changeme. Add admin to the field and select **Next**. For the password, the very first time it is changeme. You won't see it written like that. We've put it there just to make it really clear for you but it's changeme (c-h-a-n-g-e-m-e) all lowercase. Then select **Sign In**. The first thing you'll have to do is update your account by entering your email address for the admin account so put an email address in there, a confirmation email address and then select the **Change Email** button. The next thing to do we'll have to change the password for the very first time. Your existing password is changeme (c-h-a-n-g-e-m-e) and then enter your new password making sure that it matches all the requirements of how many characters...

02:00-03:00

...how many uppercase and lowercase, numbers, special characters, make sure you get a bunch of green ticks, and then confirm that password and select **Change Password**. You'll immediately be logged out of the UI. To log in with your new credentials. You can either use admin or your email. Either one of those will work and select **Next**. This will be your new password that you just created. Once you've done that, select **Sign in**. You'll be presented with the Stellar Cyber end user license agreement, the EULA, of which you can have a read. If you're happy with it, select **Accept** to move on. You'll then be presented with the default homepage of the Stellar Cyber UI. This is actually one of the visualization dashboards to select, but this is the default homepage that we'll be looking at.

03:00-04:00
The first thing to do is to convert our resource to a Data Analyzer. The first thing we need to do from the menu is select **System**. From the grouping **Data Processor**, select **Resources**. In this page you will see the resources that are configured in this Data Processor. After navigating to **System | Data Processor | Resource** page, this is what we presented with. We can see our Data Lake has already been configured and is one of the resources in this page. The other resource we're seeing is the one that we gave the host name sc-dp-da-master is available. I want you to pay particular attention to the available column and the tick meaning that resource is available for us to turn into a Data Analyzer. As the note says, do not edit the resource directly in the **Data | Data Processor | Resources** page, this page.

04:00-05:00
Instead, we're going to use the **System | Data Processor | Data Analyzer** and we're going to create a Data Analyzer utilizing this resource. Navigate to **System** under the **Data Processor** column and choose **Data Analyzer** to get to the Data Analyzer creation page. Once at this page there is a **Create** button. You can see in this screenshot there is no Data Analyzer created yet. We'll have to create one from the resource. We already know it's going to be the Data Analyzer. That's that virtual machine that we've created earlier. We open that Data Analyzer page and select the **Create** button. The pop-up menu we'll be presented with is to create our Data Analyzer. Now the example here, the name sc-dp-da-master, is exactly the same as what we call the resource.

05:00-06:00
Probably better off calling this sc-dp-da-cluster would be a better example of which the DA Master will be. The resource will be one of the devices in this cluster, the cluster of Data Analyzers. In our example, we are only going to have one Data Analyzer, just the master. We won't have any worker nodes in there. But a better example to call it the cluster rather than call the host the master and the cluster the master. That's really what we're asking here. We can see the cluster name is chosen from our Data Analyzer name. Ports are configured, but we haven't added a role yet. We haven't added a resource to that. We can see there under **Cluster Nodes** that we have an **Add Node** button. That's what we will choose next. Choose **Add Node** and then from those drop-down menus, we can see there is our Data Analyzer, our sc-dp-da-master. That's just the name of the host.

06:00-07:00
Remember we did a `set hostname`. This is where that shows up when we did that from the command line on our virtual machine. Then we have to give it a role if within this cluster. The first role

we have to have is a DA master. From that drop down, we choose **DA master**. When we've done all that, select **Submit**. We can see there, wait a few minutes as it says wait a few minutes to refresh the Data Analyzer configuration page because it has to go create that. While we're waiting there is a little recircle button just to the left of the search and we can just tap on that to refresh this screen. Once it's done, we want to make sure that the node list in the column. We've gone from zero nodes to one node. We chose a node in that last configuration screen. We also want to see that the status has a green dot in that column and that we're ready to go.

07:00-08:00
If you click on the **node** button, it will display all of the nodes within the Data Analyzer cluster. Within that Data Analyzer cluster, we have a host named sc-dp-da-master. That is our virtual machine that was created earlier in ESXi. We can see there, free disk space, plenty of disk There's the 484 which is around the era of the 500 gig we gave it in ESXi. It's all looking good there. Finally we just want to confirm a few other things. We've created our Data Lake virtual machine and configured that We've now created a resource that we've turned into a Data Analyzer. We just now want to make sure that there's a couple of bindings that have been configured correctly. The first one is to check that there's a default receiver. You can see there check default receiver configuration because Stellar Cyber automatically configures this, a default receiver...

08:00-09:00
...to be used as a destination of data collector, sensors, and other data sources. Before you add data sensors, to your new DP or new cluster, check the configuration of your default receiver. To do that we go to **System**. Under the **Collection** category, click the **Receivers** button to see what receivers we've got. There we go. We can see that there is a default receiver. Now the other piece to make sure or to confirm is that the receiver host name is set to the IP address of the DA Master, it's not set to the IP address of the Data Lake Master. Just confirm that configuration. In our example, the dl master was 192.168.1.175. We do not want to see that here. What we do want to see is the DA Master IP address of 192.168.1.150 or whatever it is for your IP addressing.

09:00-09:50
Final to check that our default sensor profiles are using the default receiver. Again, select **System**, and then from the **Collections** category, select **Sensor Profiles**. We can see there we have two default sensor profiles: default and default modular. One is a standard profile type. The other is a modular sensor profile type. But what we want to see in that Receiver column is that they're using the default_receiver that was created by default. Just a confirmation.

The lesson, *Stellar Cyber Platform - Data Processor Configuration*, is complete. Thank you.

L07B-DEMO: Data Processor Configuration (07m:30s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Stellar Cyber Platform - Data Processor Configuration. Stellar Cyber Platform - Data Processor*

Configuration. This lesson demonstrates the completion of the setup steps in the Stellar Cyber user interface.

In this demonstration, I will show you the configuration steps in the user interface, or the UI, including initial login to the UI requiring our password change, acceptance of the EULA (end user license agreement), and adding an email address. We're going to confirm the resource availability to create the Data Analyzer. Then we're going to create the Data Analyzer master. Then we're going to check on some default configuration. We're going to check on the receiver configuration.

01:00-02:00
We'll also then look to make sure that the sensor profiles have that receiver configuration in there too. Let's start with signing into the UI for the first time. Okay. We're going to start with admin, not root, admin. **Next**. changeme. Here we are again c-h-a-n-g-e-m-e. I need to update my email address. **Change Email**. Now we need to give it the original password, which was c-h-a-n-g-e-m-e (changeme) and put in our new password. Right. The fact that that came up, the button came up to change password means that they match. We've got all these five green ticks saying that I've matched the password requirements, change password. Alright.

02:00-03:00
Out we go again. Back in here. You can either use the username, which is admin, or you can use the email address that you just entered. I'm going to use admin because it's a bit quicker. Okay. This time we're asked to accept the end user license agreement. Have a read of that agreement. When you're done, select **Accept**. Here is the default home page. What we now need to go and do is confirm the resource availability. **System | Resources** and we can see a resource available. We have the host sc-dp-da-master. That's the host name we gave it in the CLI, in the console. We can see the tick there that says Available, so that's great news.

03:00-04:00
We have a resource available to turn into the DA Master. Okay. What we're going to do is **System | Data Analyzer**. We can see we do not have a Data Analyzer right here, so we need to create one. **Create**. I'm going to give it a name of Stellar Cyber-dp-da-master. That's what we're going to call it. The cluster name; it gets the cluster name from the Data Analyzer name. We can't change the ports you can see and we need to do something else in here. We need to add a node to this. We will click **Add node**. The resource where we saw we had one available. That's our host. We're familiar with that 233 IP address. It is not a DA-worker. It will be a DA-master because we need to have at least one master of course. Once that's done we select **SUBMIT**.

04:00-05:00
Now at first you'd wonder well, what happened? It says it was successful. Data Analyzer was added successfully in that green bar, but I don't see anything here. There is a button up here, the refresh button, that we can wait. We just have to wait a few moments for that to be created. So standby. Okay. We hit the refresh button. Nothing yet. No DA configuration yet. Let's standby. Let's go back and have a look at the **Resource** page while we're doing that to see what's happened over there **System | Resources.** We can see that the resource is no longer available and the status is green. But

when we go to **System | Data Analyzer** we can see we have nothing. There it is! It just appeared. It just appeared in the table. What I was going to do is go back to the console, the SSH session and show you what it looks like in there. Let's have a look in here. We can see it's now DR Master.

05:00-06:00
If we do a `show status` it's also downloading some pods. We are waiting for that. Now if we jumped over here maybe a little earlier, we would have seen that this was in a much earlier configuration and that's why it was not showing in our list. Let's wait for these all to get up and running before we go look further. What we could do in here...let's do this. We can see here in the node list that we have our sc-dp-data master. That's great. Thank you for the DA. That's what we want to see in there, but we're just going to wait for all of our pods to come up and running. There we go. Okay. We can see that we have the system ready. Let's go back to our UI. Alright. Here we are in our UI. We went in here before.

06:00-07:00
We see we do have a node in our node list, which is fantastic for our Data Analyzer configuration. But we want to also check that our default receiver and our Modular Sensor has that default receiver. Go to **System | Receivers.** We can see we have a default receiver. Let's go and have a look inside that receiver and confirm that the address is the Data Analyzer, not the Data Lake. Now when we look at that I would say that is the Data Lake's IP address, not the Data Analyzer. We want to change that to 233 to confirm that that's going to work. That's where the data is going to be received to, so click **SUBMIT**. The other thing we want to check is we want to go into our Modular Sensors. Sorry, not the Modular Sensors, the sensor profiles. They'll be two created by default: default and default modular.

07:00-07:30
One is a standard sensor and one is for Modular Sensors. We just want to go and look at those and make sure they've got that receiver configured and we can see it here actually in the column **Receivers:** default_receiver. We can see in both of those. Everything looks good. We're ready to proceed.

The lesson, *Stellar Cyber Platform - Data Processor Configuration*, is complete. Thank you.

L08 - DATA LAKE ADDITIONAL DRIVE CONFIGURATION

L08A-PRES: Data Lake Additional Drive Configuration (07m:55s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Data Lake - Data Lake Additional Drive. Data Lake - Data Lake Additional Drive*.

What we've got here is a login to the Data Lake virtual machine CLI or the console. We've logged in with our aella and our password, and we've done a `show storage` command. What we can see in the result of the `show storage` is that device named SDA one of 499.9 gig. That's our 500-gig drive

that was created when we made the virtual machine earlier. Just a 500-gig drive. Now we did create a datastore with more than 12 terabytes of data storage.

01:00-02:00

We would expect to see that in here, but at this stage the virtual machine has not had that added. Before here's a `show storage`, which shows that that does not exist. There is no significant sized drive there. No two plus terabyte in a test or a training environment. Or in a production environment there is no 12 plus terabyte storage. Let's go ahead and create that. In ESXi from the left panel under **Virtual Machines**, select our Data Lake virtual machine, our sc-dp-dl-master, and we're going to edit that virtual machine to add the additional disk. We navigate there. We select our virtual machine and from the menu above, select the **Edit** button. From the pop-up window we can see what we've already got in our virtual machine and what we need to do is add that second hard drive for the Data Lake. The datastore there is the Data Lake and create a space on there.

02:00-03:00

From above under the **Virtual Hardware** tab, we're going to select **Add Hard Disk**. Now we're going to add a standard hard disk to our virtual machine. From this pop-up window we can see our original hard drive of 500-gig and now our second hard drive that we're adding. The first thing we need to do is change the location to the datastore where we're going to store the Data Lake. We created that datastore earlier, the SC Data Lake Storage. That's the datastore that was created in ESXi where we kept adding the drives to get it to a total of more than 12 terabytes of disk space. The first thing to do is to browse to that area and choose that datastore. Once that's done we can then increase the size of our hard drive. Once you've done that...

03:00-04:00

...click up the top there and we can increase that, as you can see, to 12 terabytes. You have to make sure that there's at least 12 terabytes available to do that. In our datastore we do have that correct. Once we've browsed to the right datastore, set the disk space to the desired amount. As we said, for a production network, 12 plus terabytes of data storage in a training or a demo environment, at least two terabytes of Data Lake storage. Once that's been configured correctly, choose the **Save** button. A bit of a summary of what our hardware looks like. Our vCPUs of 40 for our Data Lake, 136-gig of RAM. The first hard drive, the 500-gig of storage is where our virtual machine is being stored. That's that use and we can see our hard disk too of now 12 terabytes that we're going to use for our Data Lake storage.

04:00-05:00

What this does is add the virtual machine. It adds the hard drive to the virtual machine, but there's more configuration steps required to change our Data Lake that is currently using the 500-gig, clearly not enough to use our 12 terabytes. From hard disk one to hard disk two we'll see what those changes look like. Now from the CLI, we can see or in the console the CLI, we can see that with a `show storage` command, there is now an sdb device name with 12T to 12 terabytes of disk storage. Fantastic! The virtual machine knows that now we added that second drive. We can see our two drives there. sda1 of 500-gig or 499.9. That's where our VM is stored and our operating system and our software. For the Data Lake we're going to add it to sdb, the 12 terabytes of storage.

05:00-06:00
The way we do that is with the `add storage blk` command. `Add storage BLK`, and then the name of the device. We saw in the previous slide the device name was sdb, so add `storage BLK SDB`. Once we enter that command we can see that add the storage will erase and format the selected disks. Make sure that we've selected the correct disks. We most definitely have. Confirm with an uppercase or a capital Y. We're looking for the result of success. Right. Now we can see that we've got mount point for our extra storage, our 12 terabytes of storage. Note that /opt/stellar/es-data-lvm. We're going to see that name when we go into the UI to migrate from the existing 499.9-gig drive or the 500-gig drive...

06:00-07:00
...which has been used for our virtual machine, our software and we will migrate that to our new data storage for our Data Lake, that 12 terabytes. Take note of that name and we'll see that again when we jump into the UI. From within the UI we're going to navigate to **System** and then to **Data Management**. Under **Data Management**, select the tab **Advanced**. From the **Advanced** tab is where we're going to do this migration. We can see there we need to migrate from the existing datastore. Where B is highlighted, there is a drop down and we will see in that drop down our new storage, our /es-data-lvm. We're going to migrate from /esdata to /es-data-lvm that was attached to.

07:00-07:55
That's the mount point for our 12-terabyte hard drive. Once you've done that, select **Start Migration**. Now that the migration is done, let's go into our Data Lake and confirm that we now have that mount point correct, and we have 12 terabytes of storage. Go to **System**, then under the **Data Processor** grouping, select **Data Lake**. Next to the node list, select that and we'll get our pop up. What we're looking for under our data storage is 12 terabytes. If we recall before, it was that 500-gigabytes of data storage. Now we're over 12 terabytes or 12,000 gigabytes. That's correct. That's what we're expecting to see.

The lesson, *Data Lake - Data Lake Additional Drive,* is complete. Thank you.

L08B-DEMO: Data Lake Additional Drive Configuration (11m:25s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module is *Data Lake - Data Lake Additional Drive. Data Lake - Data Lake Additional Drive*. This lesson includes a demonstration of adding a datastore for the Data Lake to the Data Lake virtual machine and the migration of data to this new large storage directory.

In this demonstration I will show you the configuration steps in ESXi, the Data Lake console, and the Data Processor's user interface to add the Data Lake storage to the Data Lake virtual machine. The steps will include confirmation of the datastore, and we'll look at that both in, actually we'll confirm what we first see in the UI, then we'll have a look at the datastore and confirm its existence in ESXi.

01:00-02:00
We'll validate the current storage in the Data Lake via the console. We'll add a second drive to the Data Lake virtual machine, migrate the Data Lake storage to the new drive or the new directory, and validate the additional storage and the migration was successful. Let's start with the confirmation of where we are right now. First of all, let's log into our user interface, admin. Okay. Here is our user. No data of course because we haven't configured anything. There's no Modular Sensors. There are no log forwarders. There are no connectors there. There's nothing coming into the system at the moment, so it's going to be pretty fresh. We can go to **System** and we're going to look at the Data Lake. This is the Data Lake we expect to see. You can just grab these columns to make them wider. The name is our Stellar Cyber Data Lake, our cluster.

02:00-03:00
Now, the column we're interested in is this node list. Let's make that wider so we can see it. We've got our node list right here. It's a button that we can click on. It's going to show us what is the nodes that are in our cluster. Our cluster is Stellar Cyber DL cluster (sc-dl-cluster). Inside our cluster we've got Stellar Cyber dp-dl-master (sc-dp-dl-master) in the cluster. What we're interested in here is data storage. How much data storage have we got? 484 gigabytes. Well, that's using the 500-gig that came with the virtual machine. We created that datastore that Stellar Cyber VMs datastore for the virtual machines, not for the Data Lake. We know we created another datastore for that. So let's go over to ESXi and see what's in there. We can see at the moment of our Data Lake Master we have one hard drive of 500 gig. That's it. That's where if we connected up connectors and sensors...

03:00-04:00
...log forwarders, that's where the data will be in this side there's 500 gig and that's not what we want. We know we had to have a minimum of two terabytes for our demo or training system but in a production system, 12+ terabytes of data. Let's just jump into our console and have a look at what it thinks. Alright. This is aella. Okay. Here we are. `Show storage`. We can see in here what we're seeing is that 499.9 gig, that's where it is currently storing. It's not where we want it to go. Let's have a look about adding some storage. Let's close our console. Let's confirm our datastores. We select **Storage**.

04:00-05:00
It's this one we're after–the Stellar Cyber Data Lake. 5.24 gig is certainly more than the two required for a test or a training system, but certainly not enough for a system which has a minimum of 12 gig for that. Okay. We've got that datastore and that's where we want to choose the drive to be, that we're going to add to our virtual machine. Back to **Virtual Machines**, back to our Data Lake, and select the Data Lake. We can see here we have one hard drive here of that 500 gig that matched up with the show storage we saw in the console. Let's edit. We're going to (select) the **Virtual Hardware** tab. We're going to add a hard disk. It's going to be a standard hard disk. We can see immediately we've got a hard disk one: 500 gig. There we go. Now if we look where this one is, this one is stored in our Stellar Cyber Data Lake Master, Stellar Data Lake Data Lake. Okay. Great.

05:00-06:00

If we look where this one is stored and browse. We want to make sure. Let me just expand this so you can see them all. Here are our datastores. We don't want to use the VMs because we know that's only one disk of one point seven gig but we want the big datastore the Stellar Cyber Data Lake. Okay. And select. Now we know that Data Lake was five point something gig. Why don't we just make this five? Five terabytes is how big that storage will be and we want it thick provisioned. Okay. We're going to select **Save**. Doing some work down the bottom here. Reconfiguring our VM. Complete successful. Now we can see we have hard disk one is 500 gig. Hard disk two is five terabytes. Let's go back to our console and confirm what we've got now with a `show storage`.

06:00-07:00

Okay. Now we can see we have an extra disk in here. We can see that we've got that sdb of 5T, five terabytes that didn't exist there before. But that's not enough. We need to do a little bit more. We need to add this storage. The way we do that is an `add storage blk sdb` and **Enter**. Add the storage we'll erase and format the selected disks. Yep. We know that's going to happen. So we need a capital Y to continue. It's got a bit of work to do here. Success! That's what we want to see. Let's do a `show storage` again. Now what we can see here is...

07:00-08:00

...we've got our Stellar Cyber Elasticsearch data storage, five terabytes. Take note of that /opt/stellar/es-data-lvm. We're going to see that when we go into our UI. That's the next thing we're going to do now that we've confirmed that. Okay. Back to the UI. We want to see this data storage space go from 484 to five terabytes. At the moment 484 gigabytes, we want to see five terabytes or 5,000 gigabytes. Let's do that. Okay. Let's go to **System** and **Data Management**. Under the **Advanced** tab, we'll see the current database directory is /esdata, but if we select this drop down, we'll see now is that new mount point that we saw in the `show storage` command in the CLI.

08:00-09:00

Start migration. It's going to migrate our Data Lake storage from /esdata to /es-data LVM. Select **Start Migration**. Okay. Let's log back in. Let me just jump over here and see what's happening. We can see as part of that migration that it's reset the services or the pods on our Data Lake master on our VM machine. If we do a `show status` we can see we've got nine pods running. We're expecting 39. It's not ready to log into just yet. Let's keep checking when we see our green system ready is what we're looking for. We can see we've climbed from eight pods running to 20.

09:00-10:00

Let's wait until we have the system ready before we jump back into the UI. Okay. All pods are up and running. We're just in for the Data Lake cluster to be ready. Standby. Alright. Here we go. System ready. Let's go back to our UI and sign in. When I've come back to the UI like this and there's no Stellar Cyber logo here, I know that it wasn't ready for us to sign in. There weren't some pods that had come up and running so I'm going to refresh this browser. That looks better. Let's now log in. Admin. Here we are back in our user interface. Let's go and check how our migration went, not just our migration...

10:00-11:00

...but the addition of the disk to the virtual machine and then the conversion of our space to use that disk and then our migration from the 500-gig disk to the five-terabyte disk. Let's have a look at what we can see in here. **System | Data Lake**. Our Data Lake cluster is happy. Let's go check out the node list. We can see in the node list we have data storage space of 5,079 gigabytes, which is our five-terabyte drive. Our system, our Data Processor, made up of a Data Lake and a Data Analyzer, both clustered, are ready for nodes to be added if we expand and scale this system. It is ready for use.

11:00-11:25
The next thing to do would be to add connectors, add sensors whether they're Modular Sensors, server sensors, Windows and Linux, it's ready to go.

The lesson, *Data Lake - Data Lake Additional Drive demonstration*, is complete. Thank you.

L09 - RECONFIGURING THE DATA PROCESSOR FOR CLUSTER ARCHITECTURE

L09B-DEMO: Extended Content Reconfiguring the Data Processor (33m:00s)

00:00-01:00
Reconfiguring the Data Processor is an extended content lesson in the *Stellar Cyber Open XDR Platform Installation and Configuration* module. The extended content lesson, *Reconfiguring the Data Processor,* includes a demonstration of how to reconfigure the Data Processor from a basic deployment architecture to a cluster deployment architecture.

In this demonstration, I'm going to show you the configuration steps in ESXi and the Stellar UI and the Stellar CLI on how to reconfigure the Data Processor from a basic deployment architecture to a cluster deployment architecture. The steps are going to include, like the big picture of the networking change, I'm going to show you the existing setup and we'll see that in both the UI, the CLI, and in ESXi.

01:00-02:00
We're going to shut down the Data Lake and the Data Analyzer virtual machines. We're going to add networking in ESXi. We'll be adding virtual switches and port groups, then we're going to add that networking configuration to the virtual machines–the Data Lake and the Data Analyzer–then we're going to make some console changes in the CLI to the Data Lake and the Data Analyzer around clustering. Okay. So let's start with the big picture of the networking change. We can see here this is what we've configured so far. We have a Data Analyzer and a Data Lake. They're both masters: a Data Lake master, a Data Analyzer master. They are configured in ESXi from a networking perspective to communicate with vSwitch0 through a common port group.

02:00-03:00

In this diagram the port group's called VM Network. I think we used a different port group in the lab. We'll have a look at what that is in a moment. Now what I want to show you is in this example this is the basic network architecture from a cluster network perspective. What that means is the communication that occurs between the Data Lake and the Data Analyzer occurs in the same vSwitch that communication to manage the Data Processor is the same vSwitch that information coming from the network, from our security devices and our networking devices comes in how log forwarding our network traffic, our connectors, APIs, all come in through that same virtual switch. We're going to have a look at a different way of doing that and that will be by adding a cluster or another vSwitch to it. Here's the addition. You can see here now we have this vSwitch cluster...

03:00-04:00

...down the bottom of that diagram with its own port groups. Now for the moment, let's just focus on the Data Analyzer and the Data Lake. They now have grown from having one network interface to two network interfaces, though that second network interface will be connected to this vSwitch cluster. Of course to connect to a virtual switch you have to go through a port group so we'll add a port group for the Data Lake and the Data Analyzer to communicate with that vSwitch cluster. Now in this diagram, there is a physical NIC card that is bound to that vSwitch cluster that is vmnic1. The purpose of that would be if you were to scale this deployment and have worker nodes and those workers being Data Analyzer workers or Data Lake workers and you had them on a separate physical ESXi box...

04:00-05:00

...you would have that connected to that cluster switch. Two physical hosts running a hypervisor, in this case it's running ESXi, and they need to be able to communicate. Those worker nodes need to be able to communicate with the masters. Whether they're coordinating nodes or networking nodes, they must be able to communicate. That's the reason for that external cluster switch connecting via vmnic1 to our virtual switch to be able to connect into our Data Lake and our Data Analyzer both of those being the masters in this configuration. Let's go have a look at what this looks like in ESXi, but we'll start in the UI and see what that looks like. Okay. Here we are in our UI. Let's have a look at what our system has. Okay. Probably not a lot going on here. Let's do it just a quick **Investigate | Threat Hunting**.

05:00-06:00

Turn on all of our indices to see that we do have traffic coming in. We are set to a time interval of the last 15 minutes. We can see in the last 15 minutes, yes, there has been some traffic coming in: Syslog traffic, Modular Sensor traffic, and our VMware is coming in as some Syslog traffic, but been identified via a parser port of course. Alright. We do have information coming in. Let's look at the configuration settings of our Data Lake and our Data Analyzer. We'll start by going to **System | Data Lake**. Alright. We have a Data Lake. We'll just expand this column. We have a Data Lake configuration of sc-dl-cluster. If we look at the node list to say, "How many nodes do we have participating in that cluster?" We can look and we can see we have sc-dp-dl-master.

06:00-07:00

That device is 192.168.1.234. Do pay attention to that address when we come back in and have a look later. One DL master, which has a node of one node in there at the moment. Let's have a look at the Data Analyzer. **System | Data Analyzer**, and we can see here our sc-dp-da-master if we go into our node list here's our .233. Do pay attention to this address in 192.168.1.233. We look at our node list and there is that actual host in there 192.168.1.233. I don't want you to pay attention to that because if we go back to our configuration here, this is what it currently looks like. We've got our DA Master, and the DL master, the Data Lake and the Data Analyzer masters, all sitting on the same network. That is the 192.168.1.0/24 network.

07:00-08:00
Now if we go over a look a little further, the traffic that's coming into that network is coming in via our sensor. Let's have a look at our Modular Sensor. **System | Sensors**, and we can see we have SC Modular Sensor 192. That's the name of the network it's sitting on. We can see here it is sitting with an address of 192.168.1.235. All of those devices are sitting on the same network, the 192.168.1.0 network. They're all participating in this vSwitch0, which is communicating out to the network via vmnic0. That's the current setup that we're looking at. Now is there anything else in our UI I would like to show you? I think if we just went to **Resources**, we'll just see that there is, there is our two resources with their addresses:

08:00-09:00
...the 192.168.1.233 for the DA 234 for the DL, and we saw before that our sensor was the 235. They're all like one IP address apart. Okay. Let's jump in and have a look at our ESXi box. Here is our ESXi. We can see our virtual machines. We've got our two virtual machines: our Stellar Cyber Data Lake master and our Stellar Cyber Data Analyzer master, but let's first of all go and look at the networking that's configured. We talked about that vmnic0. Here are our physical NICs. You can see vmnic0 is the only one that is connected and up and running. All the rest of them have linked down. There's a lot of networking ports on this physical box, but only one of them is up and running.

09:00-10:00
So you say, "Well, how are we going to go from having only one interface to achieving this second one?" Well, because we don't have another physical ESXi box with more virtual machines, we actually don't need this piece. We don't need this external cluster and we don't actually need this vmnic1. Really what we're going to create is just this additional virtual switch and these additional interfaces on these VM. Any communication between the Data Analyzer and the Data Lake will happen via that switch. That's what we're going to be looking at trying to achieve. Okay. That's our one physical switch. We don't have to worry about vmnic1. It's not plugged into anything, but we would of course do that if we had to scale and expand this Data Processor to a wider hardware availability with more virtual machines, more working nodes, maybe a coordinating node. We're not going to do that right now.

10:00-11:00
Let's look at virtual switches. Virtual switches: We have vSwitch0. That's the default switch that comes in ESXi and we can see we have three port groups connected to it and one uplink. Well, we just saw the uplink was that physical NIC. Let's look at the port groups that are connected to it. We have three

port groups: We have the management network, VM network, and the Stellar Cyber management interface. Going back to our diagram, this is what we've got currently right now. We did identify that it's vmnic0. And the port group is not VM Network. We're actually using Stellar Cyber Management in that, but they're all using the same virtual switch. Any communication of any device that's coming in from vmnic0 into the switch, will come into one of these three interfaces, but they're all sharing that same information.

11:00-12:00
If we want to improve upon that and have some more scalability, we're going to add this vSwitch cluster down at the bottom there. Okay. Now if we go look at our actual virtual machines, here's our Data Analyzer. You see one network card. If we look at the Data Lake, same thing: one network card. That's why it looks like this configuration here. We have a DA and a DL with one network card that is connecting to vSwitch0. Let's jump into the consoles of these devices. Here's our DA. Let's jump in there. `Aella`. What we're going to do in here is a `show interface`. We can see we only have that one interface. If we do a `show interface`, just that one interface, that management interface, 192.168.1.233.

12:00-13:00
We close that and go have a look at our Data Lake. Open up its virtual console. `aella`. Do a `show status`, show that it's all up and happy and working. Yep. Already, great. `show interface` Just the one interface `show interface all`, and still just that one interface, the 192.168.1.234. Okay. What we need to do is now go ahead. The one other thing we want to do is a `show cm` because this will be something that changes. You can see that the cm here is to .234, which is the management IP address of our Data Lake. Now if I close that and do the same thing on the Data Analyzer.

13:00-14:00
Let's open that up and do a `show cm`. We can see that it is pointing. Although the Data Analyzer here is 192.168.1.233, the cm is pointing to the management interface of the Data Lake, the 192.168.1.234. Okay. Those are some of the things...I want to show you those because they're some of the things that are going to change. Alright. Let's go and start making these changes. The first thing we need to do is we need a new virtual switch. We go to **Networking**, **virtual switches**, and add a virtual switch (select **Add standard virtual switch - New switch**). What are we going to call it? Well, let's call this vSwitch-Cluster. That's going to be our cluster switch.

14:00–15:00
Now it adds an uplink here. Uplink 1 is to vmnic1. Now that's down because as we know nothing's connected to it, we could even leave it there or we could remove it. Just to prove in this example that you don't actually even need an external interface connected to it, let's remove it. We're going to just remove that so it is just a virtual switch that's inside of this ESXi box only. Of course if we had those scalable worker nodes, coordinating nodes on a different physical ESXi box, we would need of course to add an uplink there. But let's just add it without an uplink. Bingo! We can see up here, "Virtual switch was successfully created." There it is, but nothing's connected to it. To achieve that, if we come back to our diagram that we're looking for, to achieve that so great. We've created our vSwitch cluster. Now we need a couple of interfaces here.

15:00-16:00
Well, when I say a couple, we only actually need one and both VMs will share that. Let's go ahead into port groups and create an interface. We're going to add a port group. We're going to call it...let's stick with our naming convention...Stellar Cyber, let's call it cluster. Now, it's not going to be connected to vSwitch0. We're going to connect to the virtual switch that we just created: vSwitch cluster. That's all we need to do there. What we've done so far is we've added a virtual switch vSwitch cluster. We can now see there's one port bound to it. We removed the uplinks. There's no uplink connected to it. We go to our port group and we can see our Stellar Cyber cluster is connected or bound to the virtual switch vSwitch cluster. The next thing we need to do is go and add those interfaces to our VMs.

16:00-17:00
We now need to add this Stellar Cyber cluster interface or port group, we need to add that to both of our virtual machines. We're going to go to our virtual machines. The first thing we're going to do is we're going to shut them down. Let's select them all and select **Shut down**. Alright. We can see now that the little play button has gone next to them, that they've both been shut down. Let's do the Data Lake Master first. If we look at that we can see the fact that we've got the **Power on** button, which means it's been shut down. Let's add this additional network adapter. **Edit**. We're going to add a network adapter.

17:00-18:00
That's all we have to do because it automatically bound the Stellar Cyber cluster interface to that adapter. That wasn't much to do there. Click **Save**. That was the Data Lake. Let's go to the Data Analyzer. We can see up there it was successful, good to know. We're going to go to our Data Analyzer. We're going to edit that because we can see there is only one network adapter. We are going to add a network adapter, network adapter. It's already chosen the Stellar Cyber cluster. There are other ones to choose. This is an ESXi thing. Maybe that's just the last port group that was created so it selects that to be helpful to you and definitely we want it connected when we boot this system. We've now got a DA with two network adapters.

18:00-19:00
We have a DL with two network adapters. We will now turn them on. I'm going to select both of them and power on. We can go and see are they powering on? Let's look at one of them. This is the Data Lake. Let's look at its console. We can see it's started to boot. Let's move its console over here and we'll also do the same for...I don't want to move it off the screen. Wow. That booted pretty quickly. We'll go to the DA and see where it is up to. It is also booted in this up and running. Let's start with the Data Lake. Alright. Over here to `aella`.

19:00-20:00
Then we are now if we do a `show status`, this will be in the process of loading the pods. It will certainly not be up and running. Let's at least have a look. Clearly it is busy getting those pods up and running rather than responding to my `show status` command. While we're waiting for that, let's jump in here and also...`aella` Okay. We'll do a `show status` here too. Okay. We can see here that–

wow! That was really quick for the Data Lake. All pods are running, 89 images installed. We're just waiting for the NTP and for the cluster to come up and running.

20:00-21:00
Let's see. Where do we get over here? This has got 11 pods running, expecting at least 16. 15 running. These are both going to be up and running pretty quickly. Let's just jump in and have a look. Just waiting for those to come up. I'm going to wait for them to come up before we make any further changes. We've got some configuration changes that we have to make. Let's just make sure that these come up and they're up and running and happy. Alright. There it is the Data Analyzer on the race. It came up first. Or maybe it was just the first one to show me that it was up and running first. Both of them are up and running. Okay. Let's do a `show interface`. Doesn't look much different there. `Show interface`. Doesn't look much different there but if we do a `show interface all`…

21:00-22:00
…we now have a new interface and we can see there is a link down. We've got this interface ENS 224 `show interface all`. Alright. Same thing. We have an interface, but it is currently down. Alright. We'll do the Data Lake configuration first. The first thing we're going to do is to and we're just going to come back here and just confirm what we're doing. We are telling the DA and the DL that it'll communicate via their cluster interface using that new ENS 224 interface. That's what we're going to use to communicate, which will be connected to that vSwitch cluster. That's what we're doing here. Alright. First thing to do, `set cluster_interface ENS224`.

22:00-23:00
If I've done that correctly, yes. Okay. Success! The next thing we do is we've got to give that interface an IP address. We're going to do a `set interface cluster IP` address of let's do `172.16.16`. and let's keep the address over here the same. This is `.234`. Let's make this `234/24`. An entirely different IP address. And **Enter**. Now we will do that reboot and reset, but just a couple of other commands We're going to do a `set cm` to be `172.16.16`, this address here that we just used `234`, and **Enter**. Alright. If we do a `show cm`, that's what we're expecting to see.

23:00-24:00
Now we've got to do the reboot and the reset. Let's start with that while we're waiting so reset. Sorry, no. We want to reboot first. Apologies, we want to reboot. Let that happen there. Now while that's rebooting Let's come over here to our Data Analyzer. Okay. Same thing. We've got to tell it to use this new interface as the interface to communicate as a cluster. We're going to do a `set cluster interface`–I'm pressing tab to expand those commands–to `ens224` and **Enter**. We've had success there. Fantastic. Let's give it an address. `Set interface cluster IP` of 172.16.16…

24:00-25:00
….233 to match the 233 of our 192.168/24. There's no actual reason to match but just thought we'd try and show some consistency there with a /24. We now have that done. Okay. If we do a `show cm`, the cm is no longer 192.168.1.234. The `set cm` is 172.16.16.234 as in the DL's cluster interface. And **Enter**. Alright. A `show CM` now will show a script thing. Okay. Like it asked us to do with the Data Lake, we're

going to do a reboot and a reset. `reboot`. Now while that Data Analyzer is rebooting, let's come back over here. `Aella`.

25:00-26:00
We're going to log in here `show interface all`, and we can see that we have our two interfaces. This time, our link is up with our IP addressing of 172.16.16.234, but we're going to have to wait for everything else to come up. We do a `show status`. It'll be a while for this to come up. Okay. It is still right in the pods. Okay. Nearly there: 36 of 39. `aella`. Over here on the left we're back to our Data Analyzer. We do a `show interface all`, and we can see that is up. They are able to communicate to each other. The 172.16.16.0 network we've got the Data Lake 234...

26:00-27:00
...and our Data Analyzer of 233 and they're both up and running. Alright. Let's do a `show status`. Let's see how that's going. Okay So that's still in a boot situation too. Okay I'll just wait for these to come up and running. We're just waiting for the NTP service to come up and sync. There's the Data Lake up and running. We did boot the Data Lake first, but the Data Analyzer has fewer pods to get up and running. You can see it's still got 15 running of at least 16. Okay. There you can see we now have a Data Analyzer that is system ready with all its pods up and running. Let's jump into the UI.

27:00-28:00
You were on the UI. I can't imagine that's where we last were, but let's refresh this because we have booted the device since then. Okay. Are we all up and running and happy? Let's have a look. First of all, we'll jump in here and have a look. All looks green here. We can see our machine learning services are still loading. That's sitting there at 55 percent. We'll come back and look at that. Let's have a look at our sensor. Is it up and running? Yes. Green and all good. Let's check-in **Investigate | Threat Hunting** to see that we have traffic coming in. These dips here are probably where we were doing our reboots. We can see we have traffic coming into the network. We have data coming in. Let's go ahead and have a look at our Data Lake and our Data Analyzer.

28:00-29:00
Okay. System let's look at our Data Lake first. We go to Data Lake. Alright. We have our cluster up and running happy with our 192.168.234. Nothing out of the ordinary there. We look at our node list and now we can see that we have a Data Lake of 172.16. That's how they're communicating on that 234. A bit of a change there. Now let's go ahead and look at our Data Analyzer. We can see our Data Analyzer is 192.168.1.233. We look at its node list. In there we can see its address of 172 and this is its cluster interface address of 172.16.16.233. Those interfaces are these ones that we've configured here.

29:00-30:00
We've configured one for the Data Analyzer, one for the Data Lake. Two three four for the Data Lake 233 on the 172.16.16/24 network that's communicating to our vSwitch cluster. We can see that we have adaptor 1 and adapter 2 for our Data Analyzer connected and connected. If we go look at our Data Lake, we have adaptor 1 adapter and 2 connected and connected. All is working as expected. Now if we go and look at the actual interface to show that there's traffic going across this second adapter,

this Stellar Cyber cluster connected, one of the only places we can see that is in the console or the CLI of the Data Lake or the Data Analyzer. If we come in here and do a `show interface`…

30:00-31:00
…we can see there that our cluster down at the bottom RX bytes and TX bytes has increased. This has received 1.2 gig and transmitted 40 megabytes. If we do that, do we see it grow? It's probably difficult to watch it grow from here. Well, the 40.3 has gone to 40.4. The only devices that are communicating on that network. Remember it's a virtual network. There's nothing physical connected to it except for these two virtual machines. Here we're looking at the Data Analyzer 1.2 and 40.7. Let's have a look at the interface on the Data Lake. We'll try and put it back over there where it was before so you can see them together. Line them up and line this guy up over here. We do a `show interface`.

31:00-32:00
You can see there that it sort of makes sense, right? One of them is the Data Analyzer has received 1.2 gig and the Data Lake has sent 1.1. I'm not sure exactly how the networking is counted there, sort of makes sense to me. Alright. That is it for adding. We'll go back to our PowerPoint for adding that virtual switch to an environment that was a basic setup of only having one switch. We added a second switch with additional port groups. We added those port groups to the Data Analyzer and the Data Lake. We are now in a position where we have a more scalable environment. If we were to then further scale this with worker nodes and coordinating nodes on another ESXi box, we would have to go in and add this vmnic1 back to our cluster.

32:00-33:00
Not a difficult thing to do at all there. We'd simply go back in to our virtual switch. **Networking** | **Virtual switches**. Let me close that console page. We'd go into our vSwitch cluster. Have a look at that, edit the settings, and we would add…let's just add it now. We would add an uplink. It's going to choose vmnic1. Select **Save**. That's it. Now we'll see that this should have a physical interface. Come the time that we expand and we plug into vmnic1, we would have vmnic1 able to go communicate to an external switch with another or two or three ESXi boxes to scale our Stellar Cyber platform. Thank you.

The extended content lesson, *Reconfiguring the Data Processor*, is complete. Thank you.